1. INTRODUCTION

In many organizations, technical documents are still managed through manual filing systems or scattered digital storage. These approaches depend heavily on individuals to organize and maintain records, leading to slow document retrieval, inconsistency, and a high risk of human error. Outdated versions may remain in use due to lack of proper revision tracking, and the absence of real-time access to updated documents affects efficiency, especially in fast-paced technical environments.

To address these challenges, there is a growing need for an automated and centralized system that ensures efficient document management and accurate version control. Modern technologies such as web-based frameworks, relational databases, and secure authentication systems provide an effective foundation for developing such solutions. However, generic storage systems often fail to meet the specific requirements of technical departments, such as revision traceability, structured organization, and role-based access control.

The proposed TechDoc Hub system is a web-based Technical Document Management System developed using Django (Python), along with a relational database such as SQLite/MySQL, and a Bootstrap-based user interface. The system enables users to upload, manage, and retrieve technical documents efficiently. It incorporates features such as version control, role-based access, and intelligent search to ensure that only the latest and verified documents are accessible.

The system allows users to upload documents with proper metadata including revision number, issue date, and description, ensuring complete traceability. A centralized database maintains all records and user activities, supporting audit tracking and report generation. The global search feature enables quick retrieval of documents using keywords, machine names, or categories.

Overall, TechDoc Hub provides a reliable and automated solution that reduces dependency on manual processes, minimizes errors, improves operational efficiency, and enhances safety standards in technical and industrial environments.

2. SUPPORTING LITERATURES

In recent years, significant research has been conducted on the development of specialized information systems, database management, and web-based automation for maintaining organizational accuracy and operational safety. These studies highlight the effectiveness of centralized monitoring systems in reducing human error, eliminating version confusion, and ensuring strict regulatory compliance. The following literature and technological frameworks provide a strong foundation for the proposed TechDoc Hub: A Centralized Web-Based Technical Document Management System.

2.1 Literature Review

PAPER 1:

Park, J. S., and Sandhu, R. "Role-Based Access Control on the Web." ACM Transactions on Information and System Security, Vol. 4, No. 1, 2021.

Managing security in large-scale enterprise Web environments is a critical challenge because traditional access control methods based on individual user identities do not scale effectively. In this paper, the authors proposed a comprehensive approach to managing security by marrying Web technology with Role-Based Access Control (RBAC). The study identifies two primary operational architectures: user-pull, where users retrieve and present their own roles, and server-pull, where the Web server retrieves roles directly from a central role server. To ensure security, the authors implemented these architectures using extended technologies such as secure cookies, smart certificates (X.509 extensions), and LDAP over SSL.

The study demonstrates that RBAC can significantly simplify security management by associating permissions with roles rather than individuals, ensuring the principle of least privilege. It validates that using secure, automated mechanisms to handle role information reduces the risk of unauthorized access and forgery in a way that is transparent to the user. This research directly supports the present project by providing a foundational framework for Role-Based Access Control within a web portal.

SECTION	DETAILS
Title of the Paper	Role-Based Access Control on the Web
Area of the Work	Managing and enforcing strong, efficient, and scalable security infrastructures in large-scale Web environments.
Architectures	Integration of machine vision with a real-time alerting system using strategically placed cameras and a Firebase-connected mobile app.
Methodology	Integration of RBAC models with mature web technologies including Secure Cookies, Smart Certificates (X.509 extensions), and LDAP
Security Principles	Support for Information Hiding, Least-Privilege, and Separation of Duties through role-permission.
Implementation Tools	Utilization of PGP for cryptographic routines, CGI scripts, SSL for secure transmission, and Netscape/Microsoft IIS servers

PAPER 2:

Chawan, P., et al. "Document Management System." International Journal of Trend in Scientific Research and Development (IJTSRD), Volume 2, Issue 3, 2018.

Managing mass documentation within organizations and educational institutes is a significant challenge because physical storage leads to difficulties in maintenance, updating, and rapid retrieval. In this paper, the authors proposed a web-based technological solution that transfers document functioning from traditional physical methods to a secure digital environment. The system utilizes a centralized storage method where documents are scanned and stored on a server in compressed image formats to prevent holding overloads. To address the lack of authenticity often found in digital formats, the project integrates digital attestation, where an officer applies a digital signature represented as a watermark with embedded code generated based on specific session details.

The study demonstrates that digital indexing and grouping images into folders can eliminate the tedious task of manual searching, making information available to intended users in desired time units. Security is significantly enhanced through the use of Graphical One-Time Passwords (OTP) for user login and dynamic QR codes for guest access. Guests obtain temporary access only after a user scans the dynamic QR code, which triggers the server to grant permission. This research directly supports the TechDoc Hub project by validating the transition to a centralized relational database to ensure document accuracy and long-term record maintenance. The current project extends these concepts by developing a specialized portal using Django and MySQL to manage technical documents and software versions with department-specific authorization levels driven visual analysis using deep learning to assess cleanliness levels from kitchen images, providing a simpler and more scalable solution.

SECTION	DETAILS
Title of the Paper	Document Management System
Area of the Work	Digital storage, management, and secure retrieval of documents for institutes and organizations.
Data Format	Documents stored as compressed images to optimize server performance and prevent storage overload.
Methodology	Implementation of a web-based application utilizing centralized storage and digital indexing.
Security Logic	Use of Graphical OTP for users and Dynamic QR codes based on session details for authenticating guests.
Authentication	Digital Attestation via watermarked digital signatures applied by authorized officials.

PAPER 3:

Han, J., et al. "Research on Electronic Document Management System Based on Cloud Computing." Computers, Materials & Continua, Vol. 66, No. 3, 2021.

The rapid growth of e-commerce and e-government has led to an exponential increase in electronic documents, necessitating scientific management to prevent data loss and ensure document authenticity. In this paper, the authors proposed an architecture for an electronic document management system (EDMS) built on cloud computing technology. The system leverages cloud storage and cloud security to achieve unified, reliable management, allowing users to perform multi-mode functions such as uploading, sharing, auditing, and role-based security setting. A key contribution of this work is the proposal of a secure cloud infrastructure that conducts comprehensive security services across multiple levels, including data, network, and application layers.

The study demonstrates that a cloud-based EDMS abstracts software and hardware resources into services, significantly reducing costs for organizations by eliminating the need for independent data centers. By utilizing an SOA (Service-Oriented Architecture), the system divides functions into fine-grained components, which improves overall service availability and facilitates easier business integration. This research directly supports the TechDoc Hub project by validating the use of centralized, scalable storage and the importance of standardizing document management to improve efficiency. The current project aligns with these findings by implementing a centralized web portal using Django to achieve similar goals of unified resource management and secure document retrieval.

SECTION	DETAILS
Title of the Paper	Research on Electronic Document Management System Based on Cloud Computing
Core Technology	Research on Electronic Document Management System Based on Cloud Computing
System Architecture	SOA (Service-Oriented Architecture) with a five-layer design: presentation, business, plug-in, infrastructure, and standard layers
Data Management	Unified resource allocation and load balancing for both entity data and metadata
Security Framework	Multi-level Security Cloud covering management, physical, network, system, and data security
Backup Strategy	Full backups for service code and metadata, with daily incremental backups for document libraries

2.2 Literature Review Summary

No	TITLE	FINDINGS
1	Role-Based Access Control on the Web	Identified user-pull and server-pull architectures implemented RBAC using secure cookies and "smart certificates" to solve scalability issues of identity-based access.
2	Document Management System	Proposed a centralized storage method using compressed image formats to prevent server overload integrated Graphical OTPs, dynamic QR codes, and digital attestation for secure document retrieval.
3	Research on Electronic Document Management System Based on Cloud Computing	Proposed a five-layer Service-Oriented Architecture (SOA) utilized cloud computing (IaaS/PaaS/SaaS) to reduce costs and achieve logical concentration with physical distribution of files.

2.3 Findings and Proposals

2.3.1 Findings

After reviewing the existing research papers and related works, the following findings were observed:

1. Traditional document storage methods involving scattered files and folder-based structures lead to version confusion and a high risk of utilizing outdated technical information.
2. Standard approaches to access control often rely on individual user identities, which do not scale effectively for enterprise-wide systems compared to Role-Based Access Control (RBAC).
3. The lack of structured revision tracking makes it difficult to maintain a clear history of technical document changes, affecting regulatory compliance and operational accuracy.
4. Storing mass documents as uncompressed images creates significant holding overloads on servers, necessitating centralized relational databases and optimized storage formats.
5. Research confirms that logical concentration of resources through a web portal ensures that all intended users can access verified information within desired time units.
6. While basic cloud platforms offer storage, they often lack specialized technical features like digital attestation, instrument-based navigation, or role-specific authorization for technical departments.

2.3.2 Proposal

Based on the findings from the literature survey and project requirements, the proposed system TechDoc Hub: A Centralized Web-Based Technical Document Management System is designed to address identified gaps as follows:

1. The system will utilize a Django-based web application to provide a centralized environment for managing technical documents, spare manuals, and software records.
2. Secure authentication will be implemented using official email credentials, with modification and deletion rights strictly restricted to authorized Technical Support Department (TSD) personnel.

3. Every document will be stored with a specific revision number, issue date, and revision reason, ensuring version consistency and clear audit trails.
4. An automated search engine will be developed to allow users to retrieve documents instantly using keywords, item numbers, or instrument names, reducing retrieval time.
5. All technical data and metadata will be stored in a centralized relational database (MySQL/PostgreSQL) to ensure long-term maintenance and accuracy.
6. The web interface will feature a structured, user-friendly layout with subdivisions for specific document libraries and the latest technical updates.

This proposal bridges the gap between fragmented file storage and modern digital governance, introducing an automated approach for real-time and reliable technical documentation control.

3. SYSTEM ANALYSIS

3.1 Module Description

Registration Module

- User Registration: Allows new users to create an account by providing necessary details.
- Secure Data Entry: Ensures proper validation of user information during registration.
- Access Initialization: Registers users into the system with default access permissions.

User Management Module

- User Approval: Admin can accept or reject newly registered users before granting access.
- User Control: Admin can view and manage user details.

- Account Status Management: Maintains user status such as approved or rejected.

Instrument Management Module

- Machine Management: Admin can add, update, and manage machine/instrument details.
- Machine Category Management: Allows admin to create and manage machine categories.
- Categorization: Organizes machines under specific categories for easy access.
- Structured Storage**:** Links documents to specific machines and categories for better organization.

Document Library Module

- Centralized Repository: Stores all technical documents in one place.
- File Management: Supports uploading, viewing, and downloading documents.
- Organized Storage: Maintains documents based on categories and instruments.

Announcements/Updates Module

- Information Sharing: Admin can publish announcements and updates.
- User Awareness: Displays important notices to users.
- Timely Updates: Ensures all users stay informed about system changes.

Notification Module

- Real-Time Alerts: Notifies users about updates, uploads, and announcements.
- Status Tracking: Maintains read/unread notification status.
- User Engagement: Keeps users actively informed.

Global Search Module

- Search Functionality: Allows users to search documents using keywords.
- Quick Access: Reduces time required to locate documents.
- Efficient Retrieval: Improves overall system usability.

Chat System Module

- User Communication: Enables messaging between users and admin.
- Instant Interaction: Helps resolve queries quickly.
- Message Tracking: Maintains chat history and status.

Report Module

- Report Generation: Generates reports based on system activities.
- Data Analysis: Provides insights into document usage and uploads.
- System Monitoring: Helps admin track overall system performance.

3.1 Feasibility Analysis

A project feasibility study is a critical evaluation that helps assess whether a proposed project is technically, economically, and legally viable. It examines the potential of the project to succeed within the defined scope, budget, timeline, and resources. The purpose of a feasibility study is to identify potential obstacles and ensure that the project is achievable before significant resources are committed.

3.2.1 Technical Feasibility

The TechDoc Hub system is technically feasible as it leverages a robust and industry-standard technology stack. The system uses Python with the Django framework for the backend, which is renowned for its "batteries-included" approach to security and database management. For the database, it utilizes MySQL, PostgreSQL, or SQLite, ensuring reliable relational data storage for

complex document metadata like revision numbers and issue dates. The frontend is built using Bootstrap and JavaScript, providing a responsive, instrument-based navigation interface that is accessible across various web browsers. These technologies are open-source, well-documented, and have vast community support, ensuring the system's long-term maintainability The TechDoc Hub system is technically feasible as it is developed using reliable and widely used technologies. The backend is implemented using Python with the Django framework, which provides built-in features for security, authentication, and database handling. The system uses SQLite/MySQL as the relational database to store and manage structured data such as user details, machine information, and document records.

The frontend is designed using Bootstrap, HTML, CSS, and JavaScript, ensuring a responsive and user-friendly interface that works across different devices and browsers. The system supports features like document upload, search, notifications, and chat, which are efficiently handled by the chosen technology stack.

Since all technologies used are open-source, well-documented, and easy to maintain, the system can be developed, deployed, and scaled without technical difficulty. This makes TechDoc Hub a technically feasible solution for efficient document management.

3.2.2 Economical Feasibility

The TechDoc Hub system is economically feasible as it is developed using cost-effective and open-source technologies. The project minimizes overall development and operational expenses while providing an efficient solution for document management.

- Zero Licensing Cost: The system uses Python, Django, SQLite/MySQL, and Bootstrap, which are open-source and free to use.
- Low Hardware Requirement: The application can run on standard systems without the need for high-end hardware or expensive infrastructure.

- Reduced Operational Cost: By managing documents digitally and maintaining proper version control, the system reduces errors and avoids losses caused by outdated information.
- Easy Maintenance and Scalability: The system can be easily updated and expanded without major additional costs, making it suitable for long-term use.

3.2.3 Operational Feasibility

TechDoc Hub improves the overall operational efficiency of the system by providing a structured and user-friendly platform for managing technical documents.

- Centralized Document Management: Replaces scattered storage systems with a single platform, ensuring all users access the same updated documents.
- Controlled Access: Only authorized users can upload or manage documents, reducing misuse and improving security.
- Efficient Retrieval: The global search feature helps users quickly find required documents, saving time and effort.
- Reduced Errors: Proper version control ensures that only the latest and verified documents are used.
- Improved Workflow: Simplifies daily operations and enhances productivity of users.

3.3 System Environment

The system environment defines the hardware and software requirements needed for the development and operation of the TechDoc Hub system. It ensures that the system functions efficiently and meets user requirements. The Software Requirement Specification (SRS) provides a detailed description of system functionalities, enabling developers to design and implement the system effectively while ensuring it satisfies user needs.

3.3.1 Software Environment

- **Technology Stack**: Python, JavaScript.
- **Frontend**: HTML, JavaScript, CSS.
- **Backend**: Django (Python).
- **Database**: SQLite

Development Tools

- **IDE**: Visual Studio Code (VS Code).
- **Version Control**: GitHub.
- **Database Management**: DB Browser (for SQLite).
- **Browser**: Modern Web Browsers (Chrome, Firefox, Edge).

Security & State Management

- **Authentication**: Django built-in authentication system.
- **Database Mapping**: Django ORM for handling database operations

Frontend: HTML

The Hypertext Markup Language or HTML is the standard markup language for documents designed to be displayed in a web browser. It describes the structure of a web page semantically and provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, and links. In TechDoc Hub, HTML is used to construct the instrument-based navigation and document display sections.

JavaScript

JavaScript is a light-weight object-oriented programming language used for scripting web pages. It enables dynamic interactivity, allowing users to interact directly with the portal without reloading the page every time. It is utilized here to handle the global search engine's real-time queries and dynamic UI updates.

CSS & Bootstrap

Cascading Style Sheets (CSS) is used for describing the presentation of a document written in a markup language. **Bootstrap** is specifically employed as a CSS framework to ensure a visually engaging, responsive interface that provides a consistent user experience across computers and mobile devices.

Backend: Django (Python)

Python is a high-level, general-purpose programming language that emphasizes code readability. The Django framework is used as the backbone of the system because it is a "batteries included" language that facilitates rapid development. It manages the complex logic for role-based authentication, revision tracking, and secure document uploading.

Visual Studio Code (VS Code)

VS Code is a lightweight yet powerful source-code editor developed by Microsoft. It is used for the development of TechDoc Hub due to its built-in IntelliSense, debugging tools, and integrated GitHub support, which makes managing document categories and code seamless.

SQLite

These are relational database management systems used to ensure the accuracy and long-term maintenance of all technical data. They store metadata such as revision numbers, issue dates, and user roles to enable clear traceability and strict access control.

3.3.2 Hardware Environment

Component	Specification
Hard Disk Capacity	256 GB (minimum)
RAM	8 GB
Processor	Intel Core i3 or equivalent
Display	1366 × 768 or higher

Table 3.1: Minimum Hardware Requirements

3.4 Actors and Roles

The TechDoc Hub system consists of two main actors with specific roles and responsibilities to ensure secure and efficient document management.

Admin

- System Management: Oversees and controls all system operations to ensure smooth functioning and data integrity.
- User Approval & Management: Reviews registered users and approves or rejects access to maintain system security.
- User Monitoring: Tracks user details and activities to ensure proper and authorized usage of the system.
- Machine & Category Management: Manages machines and their categories to maintain a well-structured organization of data.
- Document Lifecycle Management: Has full control to upload, update, and delete technical documents within the system.

- Version Control & Maintenance: Ensures that documents are maintained with proper versions so that only the latest files are available.
- Data Organization: Categorizes and links documents with machines and categories for easy navigation and access.
- Announcement Management: Publishes important announcements and updates to inform users about system activities.
- Notification Management: Sends and manages notifications related to approvals, updates, and document activities.
- Search Optimization: Ensures documents are properly organized to support efficient search functionality.
- Report Management: Generates reports to monitor system usage and document-related activities.
- System Monitoring: Continuously observes system performance and ensures efficient operation.

User

- User Registration: Registers into the system by providing required details for account creation.
- Secure Login: Accesses the system using valid credentials after admin approval.
- Profile Access: Views and manages basic personal account information.
- Document Access: Views available technical documents uploaded in the system.
- Document Download: Downloads required documents for reference and use.
- Search Functionality: Uses the global search feature to quickly find documents using keywords.
- Category-Based Navigation: Browses documents based on machine categories for easier access.
- Machine-Based Access: Accesses documents related to specific machines or instruments.
- Notification Access: Receives and views notifications about system updates and activities.
- Announcement Viewing: Stays informed by viewing announcements published by the admin.
- Chat Communication: Communicates with admin or other users through the chat system.
- System Usage: Interacts with the system regularly to access and utilize available resources.

4. SYSTEM DESIGN

4.1 Use case model

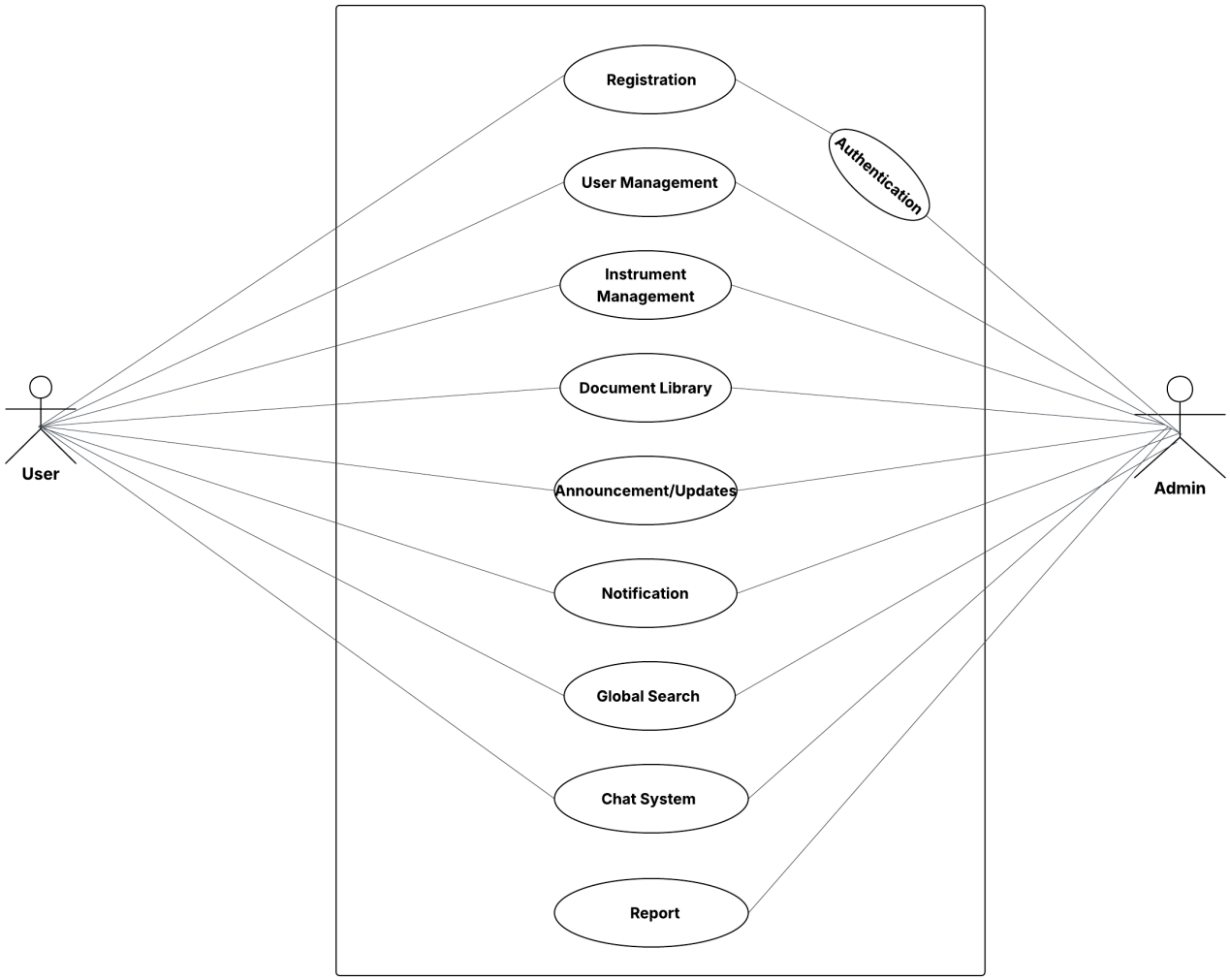


Figure 4.1: Use Case Model

The use case diagram for the Technical Document Management System illustrates the interactions between the system's primary actors the Admin and the User and its core functionalities.

The Admin, overseeing the entire system, handles user and access management through official email authentication and Role-Based Access Control. They maintain full authority over the content lifecycle, including the uploading, modification, and deletion of technical documents, spare manuals, and software records. Furthermore, the Admin ensures traceability by managing version control with revision numbers and reasons, categorizes information into logical modules, and broadcasts technical updates via the dashboard while monitoring the database for operational reports. The User interacts with the system through secure authentication to access the document library. They utilize the global search engine and instrument-based navigation to find specific data by keywords or categories. Finally, Users consume critical knowledge by viewing or downloading the most recent verified manuals and software records, while monitoring the dashboard for real-time awareness of technical changes.

4.2 Activity Diagram

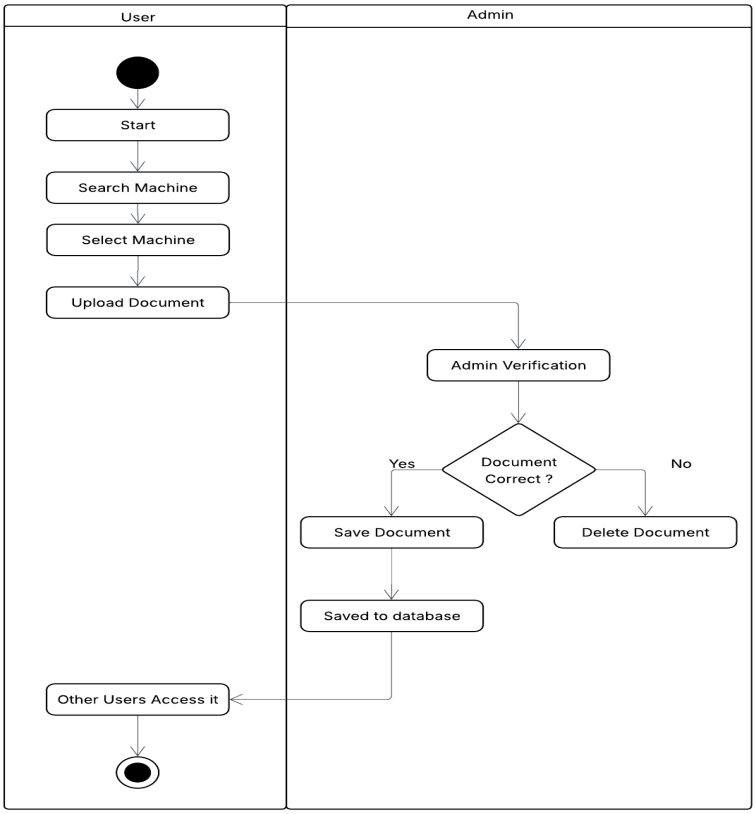


Figure 4.2 Activity Diagram of File upload

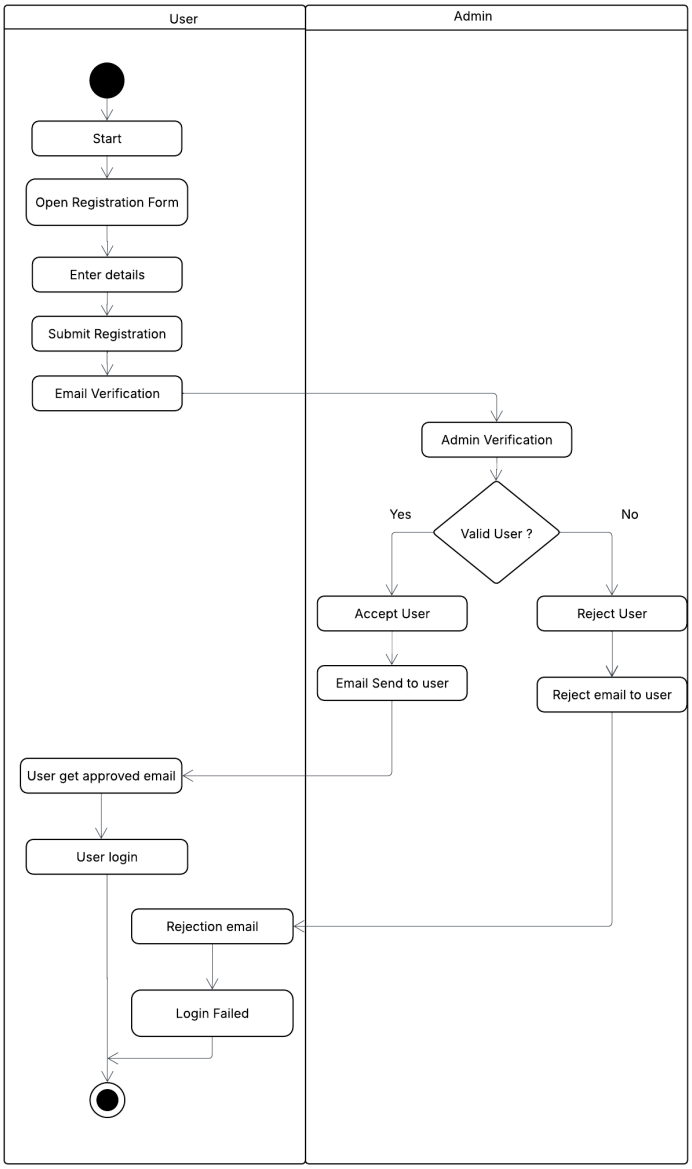


Figure 4.2 Activity Diagram of User registration

This flowchart represents the activity flow of the TechDoc Hub Web Portal Knowledge Management System, starting from user login, where users are classified as Admin, User. The Admin is responsible for overall system oversight, including user management, access control through role-based authentication, database maintenance, and monitoring system performance to ensure secure and uninterrupted operation. Admin focus on content-related operations such as uploading, editing, and deleting technical documents, assigning version details, categorizing

documents into appropriate modules, and publishing announcements to keep the repository accurate and up to date. Users interact with the system primarily to authenticate using official credentials, search or browse documents through intelligent search and instrument- based navigation, access and consume technical knowledge, and view dashboard updates and announcements. The diagram clearly illustrates the distinct workflows and responsibilities of each role, showing how their activities are coordinated within the system from login to completion of tasks.

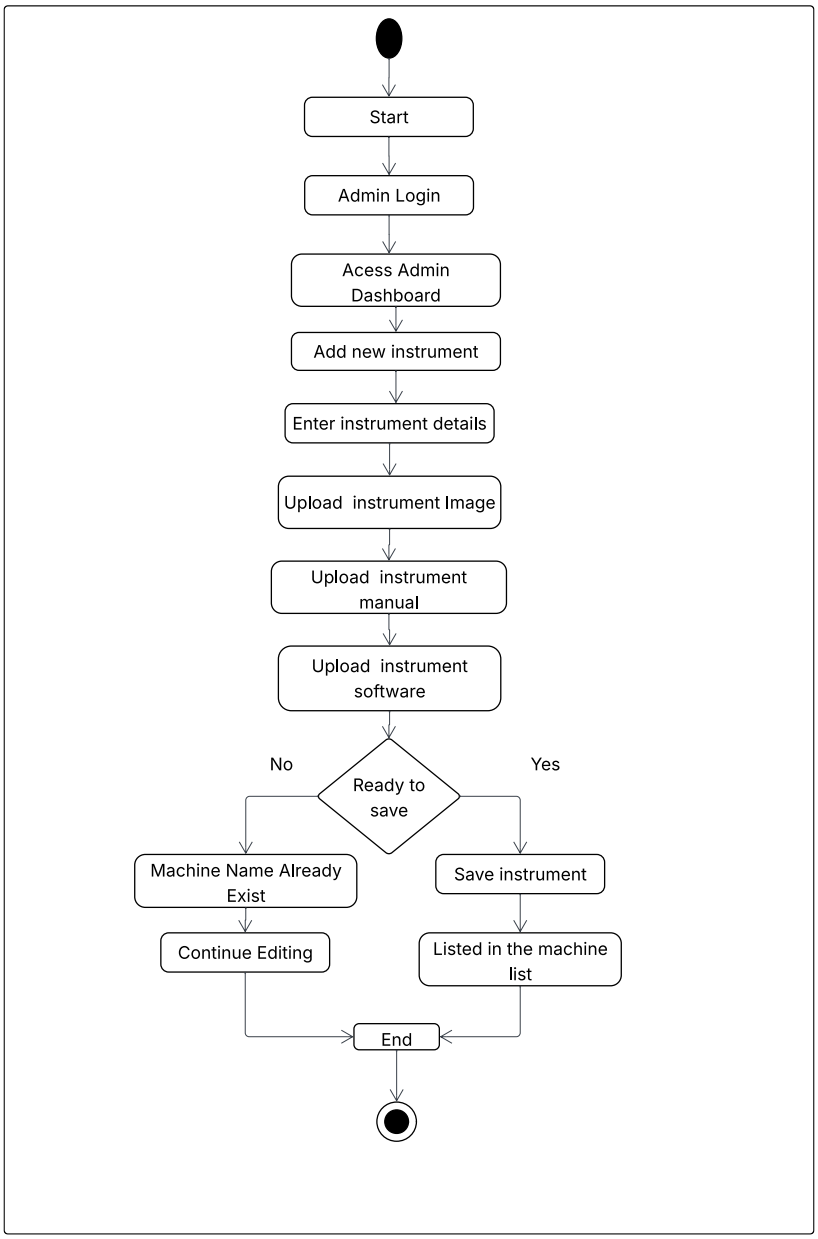


Figure 4.3 Activity Diagram of Add Machine

The Activity Diagram for the TechDoc Hub Web Portal Knowledge Management System outlines the specific sequence of actions required for an Administrator to populate the digital repository with new technical resources. The process initiates with a secure Admin Login, granting access to the specialized Admin Dashboard, which serves as the central hub for system oversight. From this interface, the Administrator triggers the Add new instrument workflow, a structured data-entry sequence that begins with entering specific instrument details. This is followed by a multi-step media and documentation upload phase, where the Admin sequentially attaches an instrument image, the corresponding technical manual, and the necessary instrument software to the record.

The workflow then reaches a critical decision point labeled Ready to save, which acts as a validation gate for the system's data integrity. If the system detects a conflict, such as the Machine Name Already Existing in the database, the Admin is redirected to Continue Editing to resolve the duplication. Conversely, if the entry is unique and the validation is successful, the Admin can Save instrument, effectively committing the new record to the database. This successful path concludes with the instrument being officially Listed in the machine list, making the technical knowledge available for General Users to search and consume, before the activity reaches its final termination point.

4.3 Sequence Diagram

4.3.1 Document Search Flow

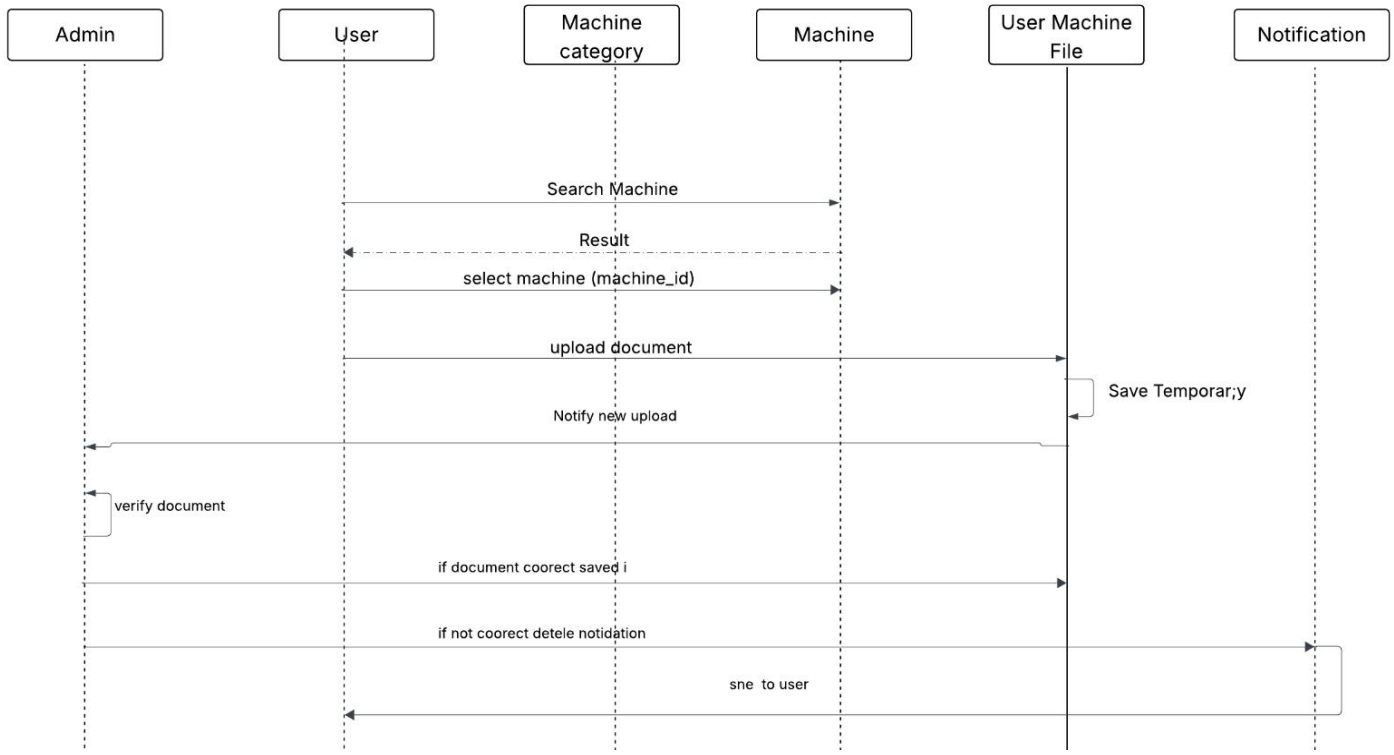


Figure 4.3 Sequence Diagram of Add Machine File

The Sequence Diagram for the Add Machine File process outlines the collaborative interactions between the User, Admin, and system modules to ensure the accuracy of uploaded technical data. The workflow begins when a User initiates a Search Machine request to the Machine entity, which returns the relevant Result. Upon identifying the correct equipment, the User selects a specific machine_id and proceeds to upload document, which is initially directed to the User Machine File for a Save Temporary action. Once stored temporarily, the system sends a Notify new upload alert to the Admin, prompting a formal verify document procedure.

The final stage of the sequence is determined by the Admin's validation results, creating two distinct logical paths. If the document is deemed correct, it is permanently saved to the machine file system to ensure the repository remains reliable and up to date. Conversely, if the document is not correct, the Admin triggers a delete notification to clean up the temporary entry. In either scenario, the Notification system communicates the final status by sending a message to user, thereby completing the feedback loop and ensuring the user is informed of the document's verification status.

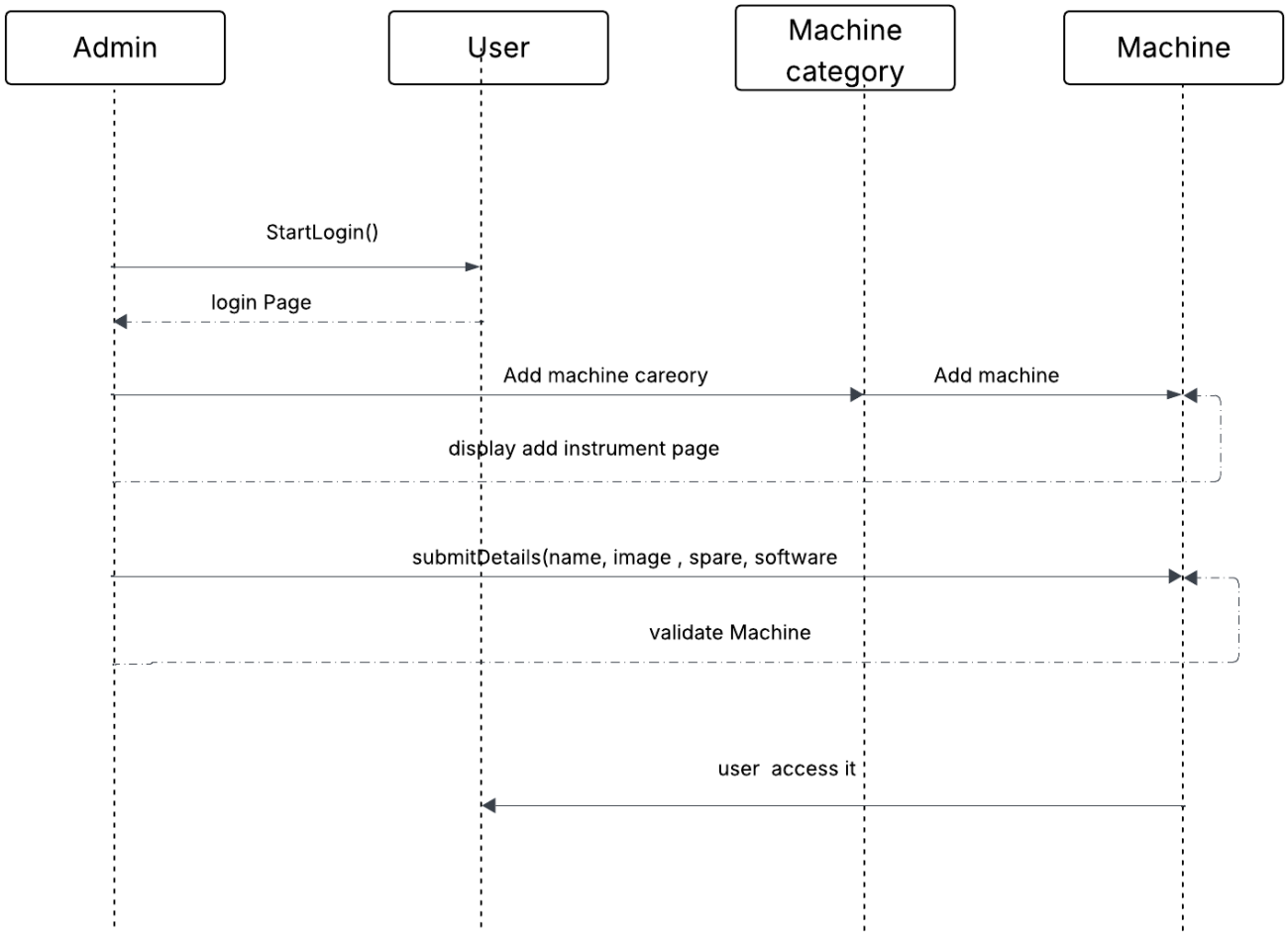


Figure 4.3 Sequence Diagram of Add Machine

The Sequence Diagram of Add Machine illustrates the chronological interactions between the Admin, User, and the machine-related modules during the instrument creation process. The workflow initiates with a StartLogin() request from the Admin, which prompts the system to return the login Page for authentication. Once logged in, the Admin triggers the Add machine category action, which cascades into the Add machine command within the backend. In response, the system generates and displays the add instrument page, providing the interface necessary for data entry.

The process continues as the Admin uses this interface to submitDetails, including the machine's name, image, spare parts info, and software records directly to the Machine entity. Following the submission, the system performs an internal validate Machine check to ensure data consistency and prevent duplication. Upon successful validation, the workflow concludes by enabling user access, effectively bridging the gap between the administrative setup and the end-user's ability to utilize the new machine record within the portal.

4.4 List of identified classes, attributes, and their relationships

Classes play a crucial role in object-oriented design by defining the attributes, methods, and interactions between different entities within the system. Below is the list of identified classes and their relationships in the document management system.

4.4.1 Identified Classes

1. Admin
2. User
3. Machine Category
4. Machine
5. Announcement
6. User Machine File
7. Notifications
8. Chat Message

4.4.2 Identified Attributes

The Admin class in the TechDocHub system is designed to oversee and manage the key administrative functionalities of the platform. It contains attributes such as admin_id, name, contact, employeid, email, gender, photo, and password, which uniquely identify each administrator and ensure secure access to the system by storing essential personal and login information for proper authentication and communication within the platform. The Admin class also provides several methods that enable the administrator to efficiently manage and control system operations, including adding machine categories, registering new machines, publishing announcements, viewing user uploaded machine files, and sending notifications to users. Through these functionalities, the administrator can maintain organized system data, monitor user activities, and manage information related to machines and uploaded files, ensuring smooth platform operation, improved system reliability, and timely updates for users regarding machines, announcements, and other important notifications.

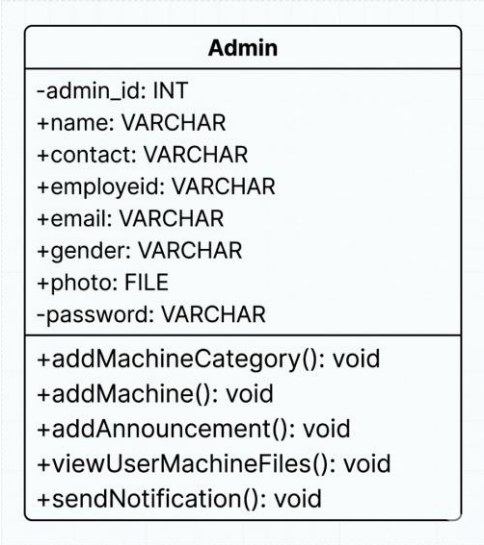


Fig 4.8 Admin Class

The User class in the TechDocHub system serves as the main structure for individuals who access and interact with the platform. It includes important attributes such as user_id, name, contact, employeid, email, gender, photo, password, and status, which help uniquely identify each user and provide secure authentication for accessing the system while storing the user's personal details, login credentials, and account status. The User class provides several core functionalities that allow users to interact with the system efficiently, including register() and login() for creating accounts and securely accessing the platform, and updateProfile() for modifying personal information such as contact details or profile photos. In addition, users can upload machine files, view machine details, read announcements, receive notifications, and communicate with other users through chat messages, allowing them to actively participate in the platform. These attributes and functionalities enable smooth interaction between users and the system while ensuring proper data management, communication, and access to machine-related information within the platform.

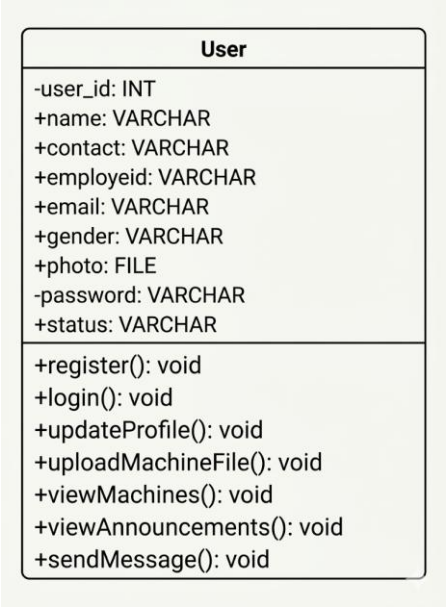


Fig 4.9 User Class

The Machine Category class in the TechDocHub system is responsible for organizing machines into different categories within the platform so that machine information can be structured and easily accessed by users. It includes attributes such as category_id and category_name, which uniquely identify each category and store the name of the machine category in the system. These attributes help maintain a well-organized classification of machines and allow each machine to be associated with a specific category. The Machine Category class also provides several methods that allow administrators to manage the categories effectively, including addCategory(), updateCategory(), deleteCategory(), and viewCategories(), which enable the administrator to create new categories, modify existing ones, remove unnecessary categories, and view all available machine categories. These functionalities ensure proper organization of machines within the system and help users easily find machines based on their categories.

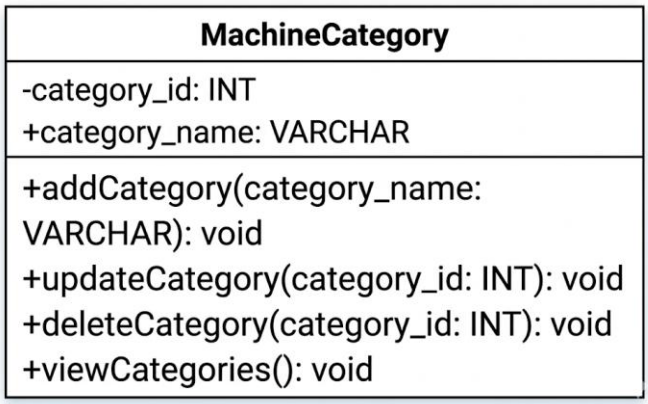


Fig 4.10 MachineCategory Class

The Machine class in the TechDocHub system is responsible for storing detailed information about the machines available on the platform. It includes attributes such as machine_id, category, machine_name, description, image, spare_manual, and software_file, which help identify each machine and maintain its related information in the system. The category attribute links each machine to a specific machine category, allowing the system to organize machines efficiently. The machine_name and description attributes store the basic details of the machine, while image provides a visual representation of the machine. Additionally, spare_manual and software_file allow the system to store important documents such as machine manuals and related software files. The Machine class also provides methods such as addMachine(), updateMachine(), deleteMachine(), and viewMachineDetails(), which allow administrators to manage machine information effectively and ensure that users can easily access machine details and related documents within the platform.

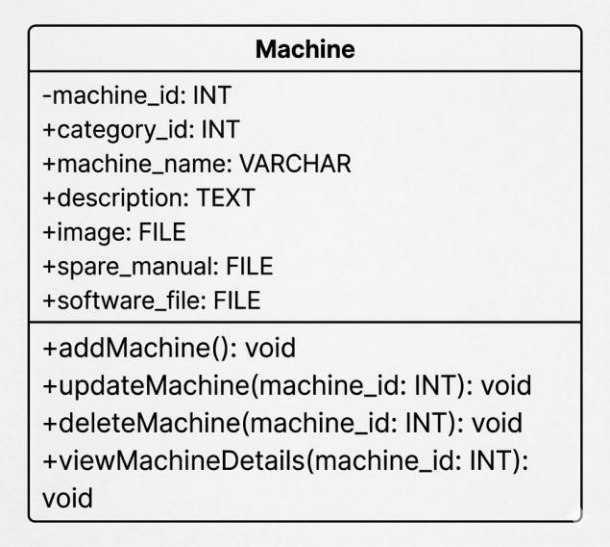


Fig 4.11 Machine Class

The Announcement class in the TechDocHub system is responsible for managing important messages or updates that are shared with users on the platform. It includes attributes such as announcement_id, title, description, and created_at, which help uniquely identify each announcement and store its content and creation time. The title attribute stores the heading of the announcement, while the description contains the detailed information or message intended for users. The created_at attribute records the date and time when the announcement was published.

This class enables administrators to keep users informed about system updates, new machines, important notices, or other relevant information. It also provides methods such as addAnnouncement(), updateAnnouncement(), deleteAnnouncement(), and viewAnnouncements(), allowing administrators to manage announcements effectively and ensure that users always receive the latest updates from the platform.

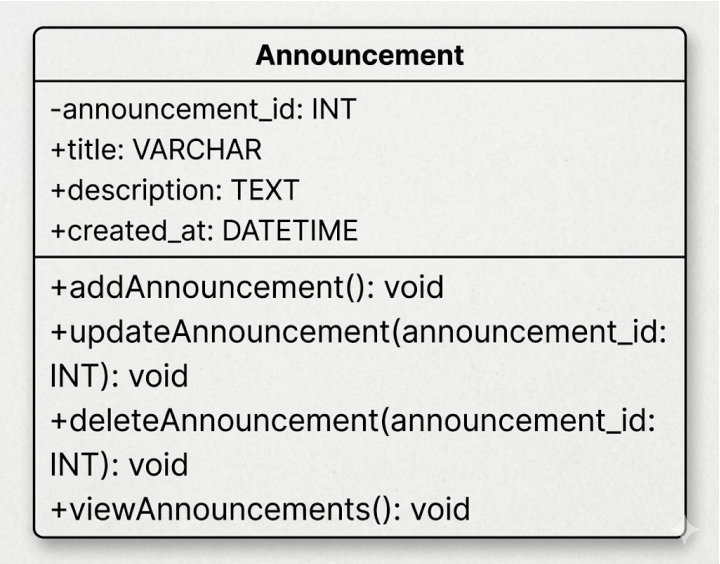


Fig 4.12 Announcement Class

The User Machine File class in the TechDocHub system is responsible for storing and managing files uploaded by users that are related to specific machines. It includes attributes such as file_id, user, machine, file, and upload_date, which help track the uploaded files and the users responsible for them. The user attribute connects the file to the user who uploaded it, while the machine attribute links the file to the specific machine it belongs to. The file attribute stores the uploaded document, which may include manuals, technical files, or other machine-related documents. The upload_date attribute records the date and time when the file was uploaded. This class enables users to contribute useful machine-related files to the platform and allows administrators to monitor and manage these uploaded documents. Methods such as uploadFile(), deleteFile(), and viewFiles() help manage the uploading, removal, and viewing of machine-related files within the system

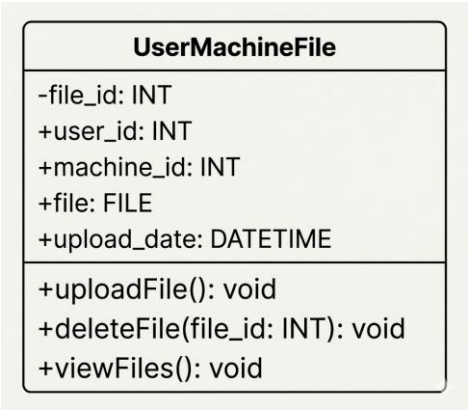


Fig 4.12 UserMachineFile Class

The Notification class in the TechDocHub system is responsible for managing alerts and messages sent to users within the platform. It includes attributes such as notification_id, user, message, and created_at, which help uniquely identify each notification and store the message information along with the time it was created. The user attribute links the notification to a specific user who will receive the message, while the message attribute contains the content of the notification. The created_at attribute records the date and time when the notification was generated. This class allows the system or administrators to inform users about important updates such as announcements, machine information, file uploads, or other system activities. It also provides methods such as sendNotification(), viewNotifications(), and deleteNotification(), which help manage and deliver notifications efficiently within the platform.

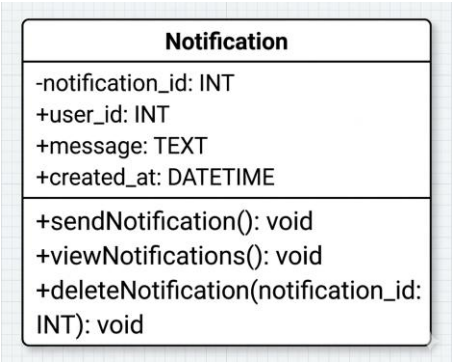


Fig 4.13 Notification Class

The Chat Message class in the TechDocHub system is responsible for managing communication between users within the platform. It includes attributes such as message_id, sender, receiver, message, sent_time, and status, which help uniquely identify each chat message and store the details of the conversation. The sender attribute represents the user who sends the message, while the receiver attribute represents the user who receives it. The message attribute stores the text content of the chat, and the sent_time attribute records the date and time when the message was sent. The status attribute indicates the state of the message, such as whether it has been sent or seen. This class enables users to communicate with each other regarding machines, uploaded files, or other platform-related discussions. It also provides methods such as sendMessage(), viewMessages(), and deleteMessage(), which help manage and control user conversations within the system.

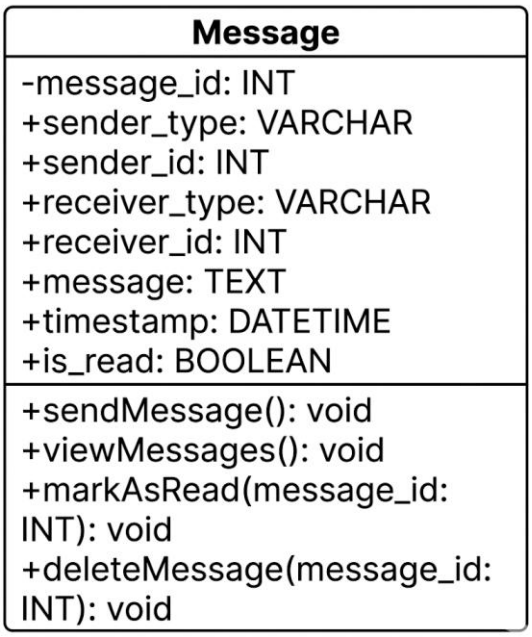


Fig 4.13 Message Class

4.4.3 Identified Relationship

Class	Related Class	Association Name	Cardinality
Admin	User	Authenticates and manages user accounts	1 - *
Admin	MachineCategory	Manages machine categories	1 - *
Admin	Machine	Adds and manages machines	1 - *
Admin	Announcement	Publishes announcements	1 - *
Admin	Notification	Sends notifications to users	1 - *
Admin	UserMachineFile	Monitors and deletes uploaded files	1 - *
User	UserMachineFile	Uploads machine related files	1 - *
Machine	UserMachineFile	Receives uploaded files	1 - *
MachineCategory	Machine	Categorizes machines	1 - *
User	Notification	Receives notifications	1 - *
User	Message	Sends messages	1 - *
User	Message	Receives messages	1 - *

Table 4.1: Identified Relationships Department

4.5 Class Diagram

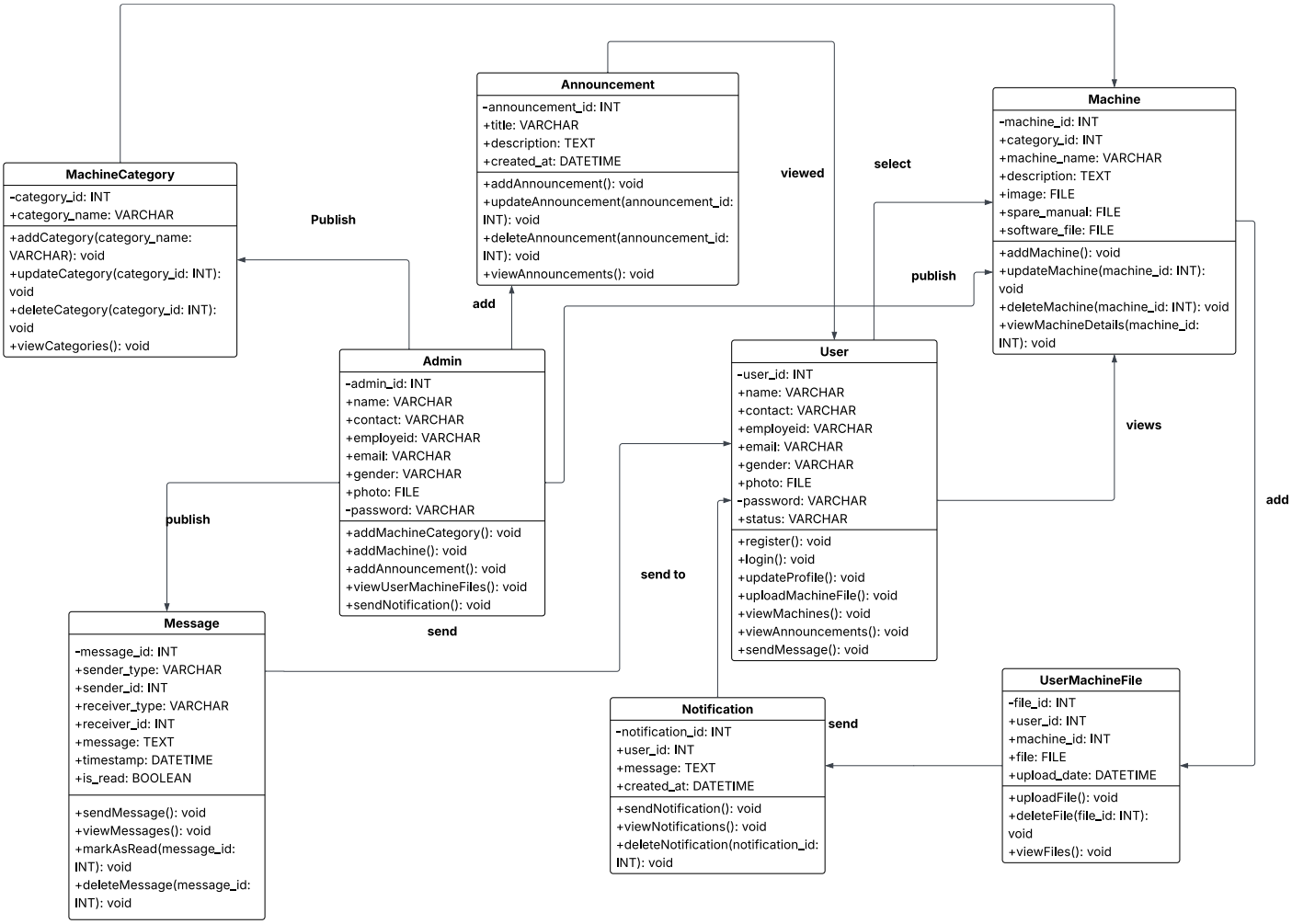


Fig 4.8 Class Diagram

4.6 Database Design

Table name : Users

Primary Key: user id

Attribute	**Datatype**	**Size**	**Description**	**Constraints**
user_id	Integer	Unsigned	Unique identifier for each user	Primary Key
name	Varchar	50	Name of the user	Not Null
contact	Varchar	50	Contact number of the user	Not Null
employeid	Varchar	50	Employee ID of the user	Not Null
email	Varchar	100	Email address of the user	Unique Key
gender	Varchar	50	Gender of the user	Not Null
photo	Varchar	255	Profile photo of the user	—
password	Varchar	50	Password used for login	Not Null
status	Varchar	20	Status of the user account	Default: pending

Table 4.2: Users Table

Table name : Admin

Primary Key: admin id

Attribute	**Datatype**	**Size**	**Description**	**Constraints**
admin_id	Integer	Unsigned	Unique identifier for each administrator	Primary Key
name	Varchar	50	Name of the administrator	Not Null
contact	Varchar	50	Contact number of the administrator	Not Null
employeid	Varchar	50	Employee ID of the administrator	Not Null
email	Varchar	100	Email address of the administrator	Unique Key
gender	Varchar	50	Gender of the administrator	Not Null
photo	Varchar	255	Profile photo of the administrator	Nullable
password	Varchar	50	Password used for admin login	Not Null

Table 4.3: Admin Table

Table name : Machine Category

Primary Key: category_id

Attribute	**Datatype**	**Size**	**Description**	**Constraints**
category_id	Integer	Unsigned	Unique identifier for each machine category	Primary Key
category_name	Varchar	100	Name of the machine category	Not Null

Table 4.4: Machine Category Table

Table name : Machine

Primary Key: machine_id

Foreign Key : category_id

Attribute	**Datatype**	**Size**	**Description**	**Constraints**
machine_id	Integer	Unsigned	Unique identifier for	Primary Key

			each machine	
category_id	Integer	Unsigned	Reference to the machine category	Foreign Key
machine_name	Varchar	100	Name of the machine	Not Null
description	Text	—	Description or details of the machine	Not Null
image	Varchar	255	Image file of the machine	Nullable
spare_manual	Varchar	255	Manual file related to machine spare parts	Nullable
software_file	Varchar	255	Software file related to the machine	Nullable

Table 4.5: Machine Table

Table name : Announcement

Primary Key: announcement_id

Attribute	**Datatype**	**Size**	**Description**	**Constraints**
announcement_id	Integer	Unsigned	Unique identifier for each announcement	Primary Key
title	Varchar	200	Title of the announcement	Not Null
description	Text	—	Detailed information about the announcement	Not Null
created_at	Datetime	—	Date and time when the announcement was created	Not Null

Table 4.5: Announcement Table

Table name : User Machine File

Primary Key: file_id

Foreign Key : user_id, machine_id

Attribute	**Datatype**	**Size**	**Description**	**Constraints**
file_id	Integer	Unsigned	Unique identifier for each uploaded file	Primary Key
user_id	Integer	Unsigned	Reference to the user who uploaded the file	Foreign Key
machine_id	Integer	Unsigned	Reference to the machine related to the file	Foreign Key
file	Varchar	255	File uploaded by the user related to the machine	Not Null
upload_date	Datetime	—	Date and time when the file was uploaded	Auto Generated

Table 4.6: User Machine File Table

Table name : Notification

Primary Key: notification _id

Foreign Key : user_id

Attribute	**Datatype**	**Size**	**Description**	**Constraints**
notification_id	Integer	Unsigned	Unique identifier for each notification	Primary Key
user_id	Integer	Unsigned	Reference to the user who receives the notification	Foreign Key
message	Text	—	Content of the notification message	Not Null
created_at	Datetime	—	Date and time when the notification was created	Auto Generated

Table 4.6: Notification Table

Table name : Message

Primary Key: message_id

Foreign Key : sender_id, receiver_id

Attribute	**Datatype**	**Size**	**Description**	**Constraints**
message_id	Integer	Unsigned	Unique identifier for each chat message	Primary Key
sender_id	Integer	Unsigned	Reference to the user	Foreign Key

			who sends the message	
receiver_id	Integer	Unsigned	Reference to the user who receives the message	Foreign Key
message	Text	—	Content of the chat message	Not Null
sent_time	Datetime	—	Date and time when the message was sent	Auto Generated
status	Varchar	20	Status of the message (sent / seen)	Default: sent

Table 4.6: Chat Message Table

4.7 UI Design

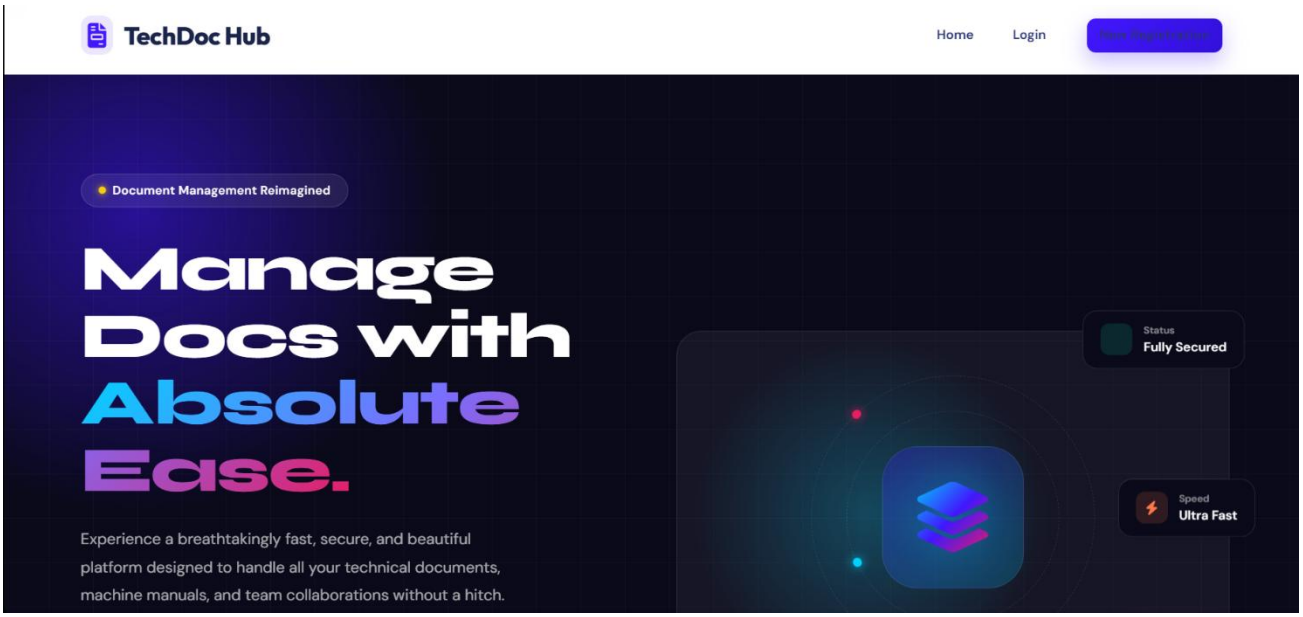


Figure 4.24: Home Page of TechDoc Hub

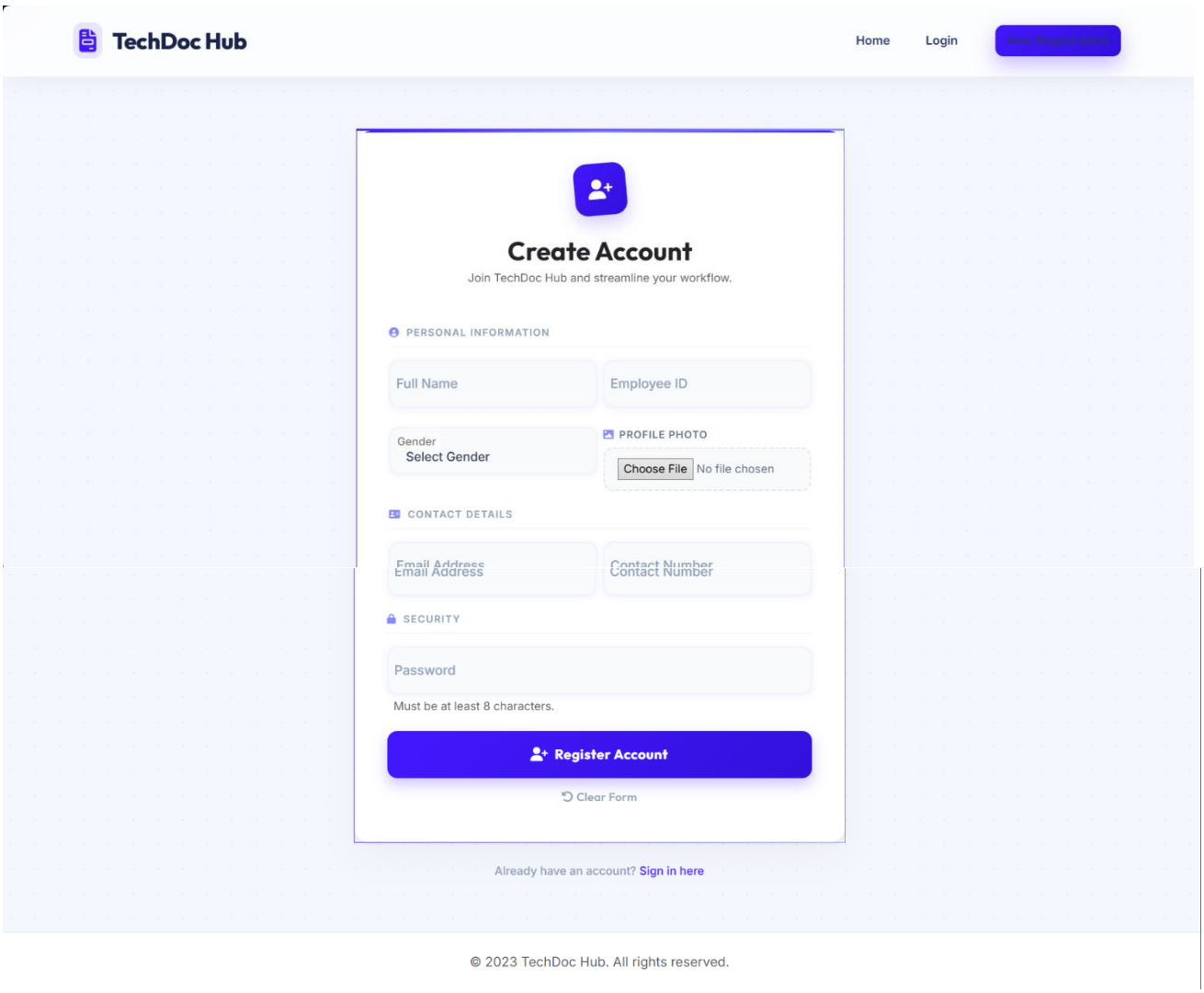


Figure 4.24: Registration Page of TechDoc Hub

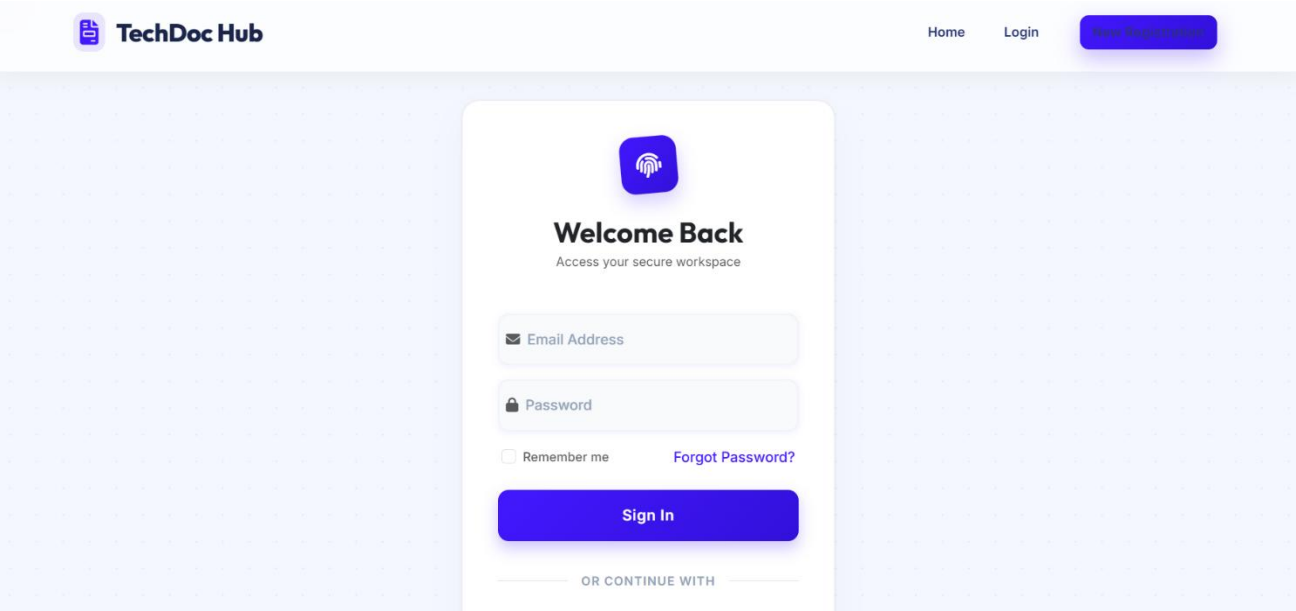


Figure 4.25: Login Page of TechDoc Hub

The above three figures depict the Homepage ,User Registration Form and Login Interface of the TechDocHub system. The registration form allows new users to create an account by entering details such as name, email, contact number, employee ID, gender, password, and profile photo. After successful registration, users can access the login interface, which enables them to sign in to the system using their email and password. These interfaces ensure secure user authentication and allow registered users to access the platform to view machine information, upload machine-related files, and interact with other users through the system.

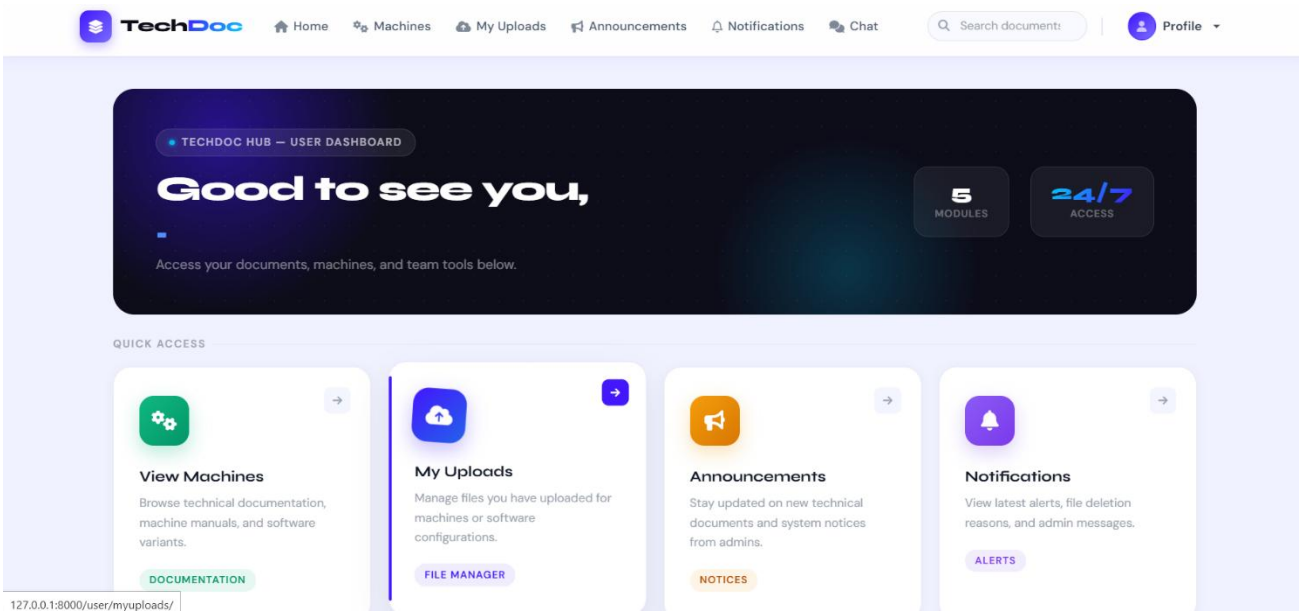


Figure 4.26: User Home Page

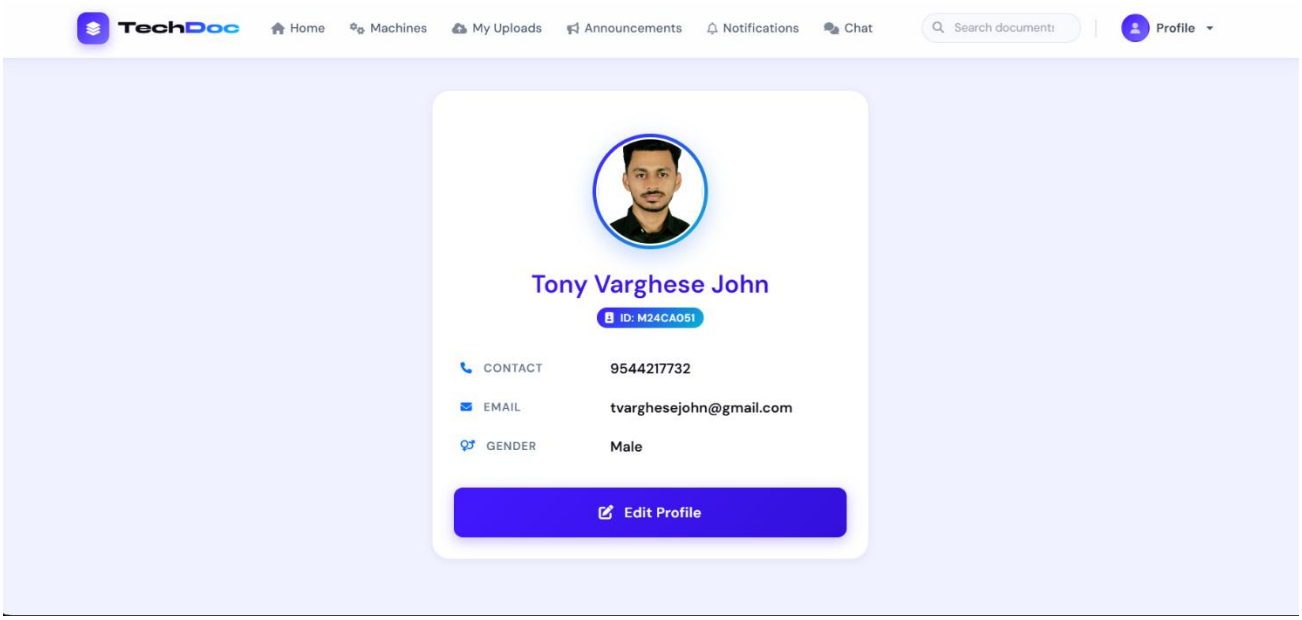


Figure 4.27: User Profile Page

The homepage of the TechDocHub system provides users with an organized interface to explore different machine categories and view detailed information about machines, including descriptions,

images, manuals, and software files. It also displays important announcements from the administrator to keep users updated with system activities. The My Profile page shows the user's personal details such as name, contact number, employee ID, email, gender, and profile photo, allowing users to view and manage their information while interacting with system features like uploading machine files and viewing machine details.

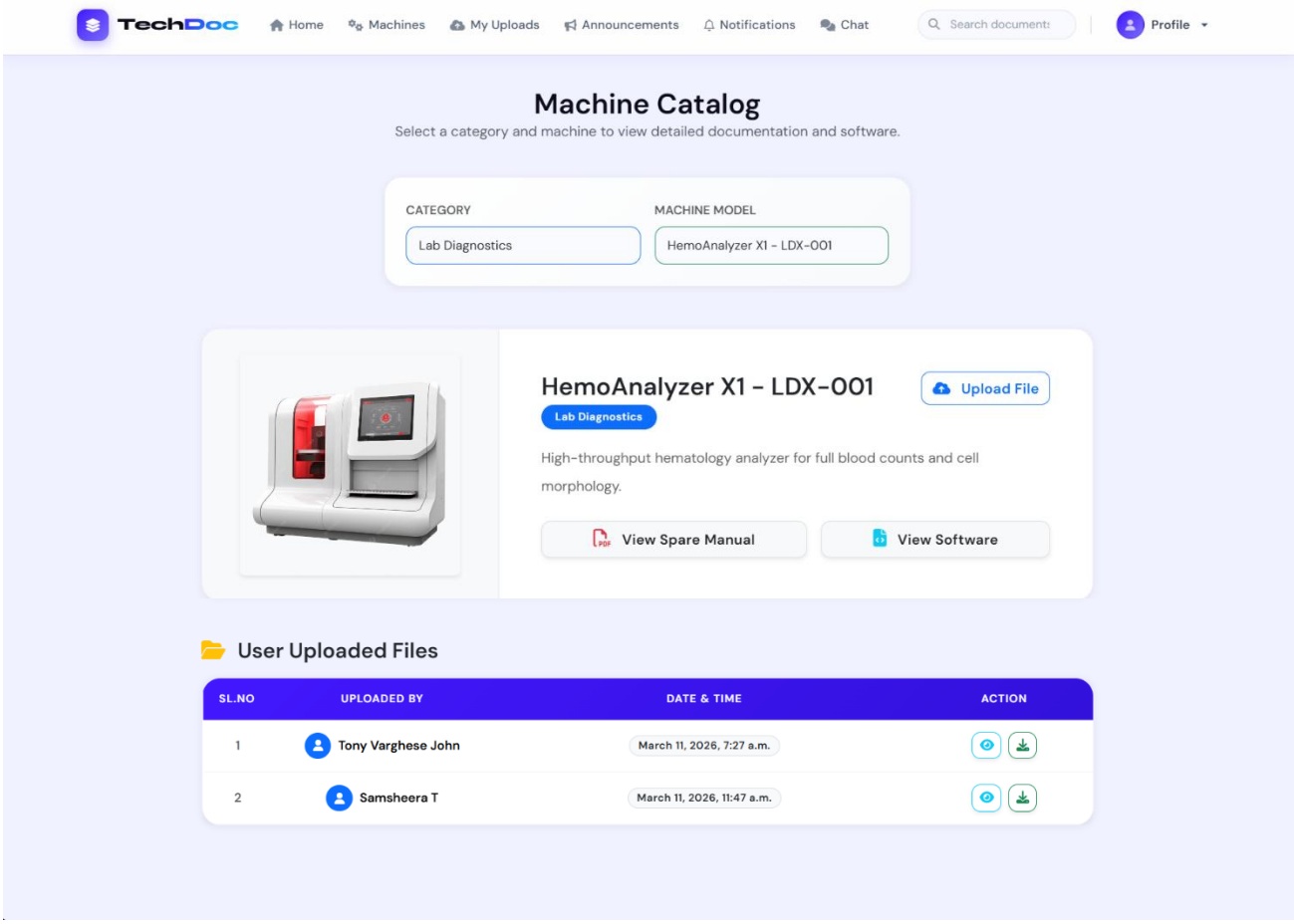


Figure 4.28: Select machine

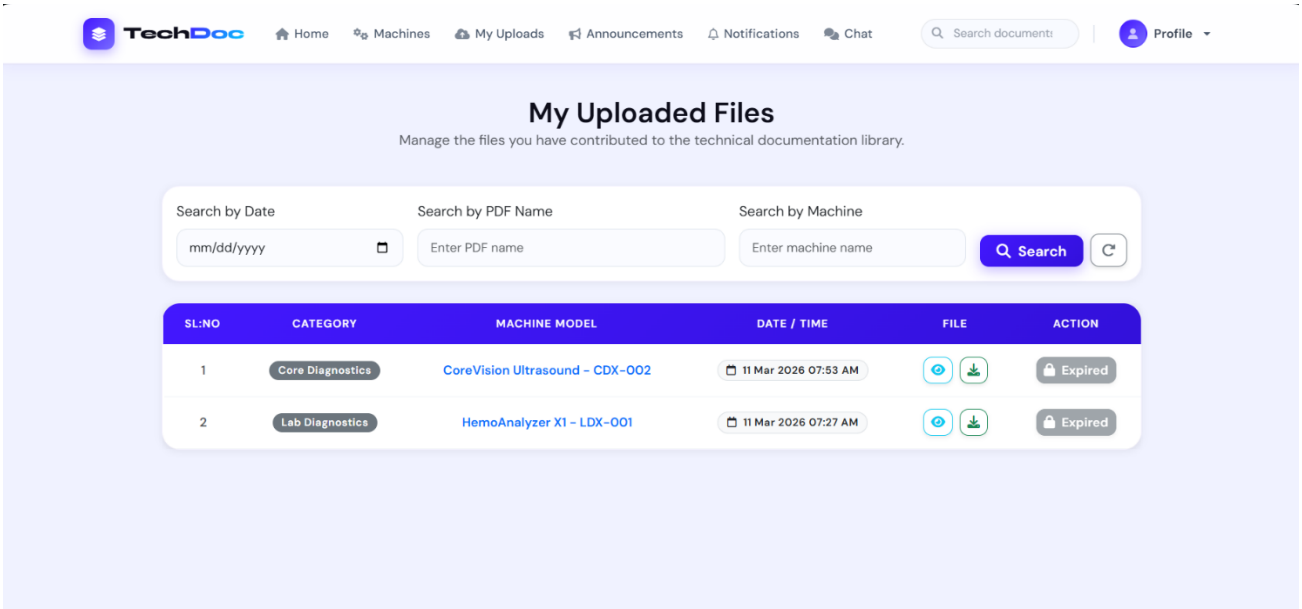


Figure 4.28: My Uploads

The two images showcase the user interface of the TechDocHub web application. The first image displays the Machine Selection and File Upload page, where users can choose a machine from the available list and upload related files such as manuals, documents, or software files. The page provides a simple interface for selecting the machine and attaching the required file, allowing users to easily contribute machine-related resources to the platform. The second image shows the My Uploads page, which displays the files uploaded by the user along with the corresponding machine details and upload information. This section allows users to view and manage their uploaded files, helping them keep track of the documents they have shared on the platform.

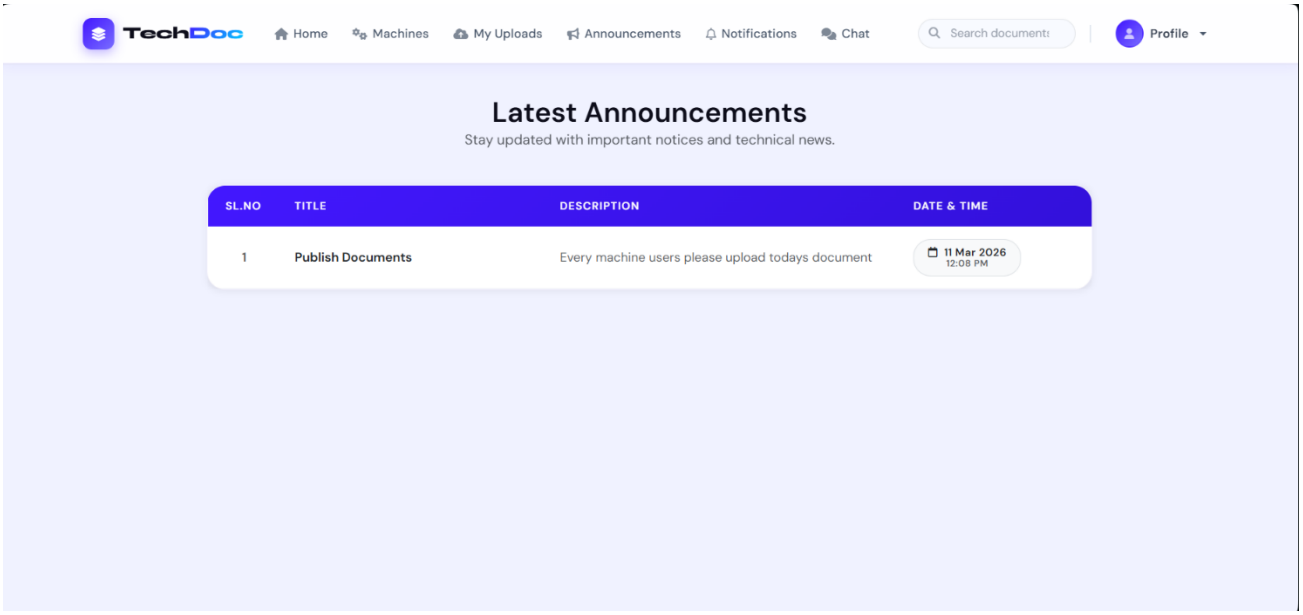


Figure 4.29: Announcements

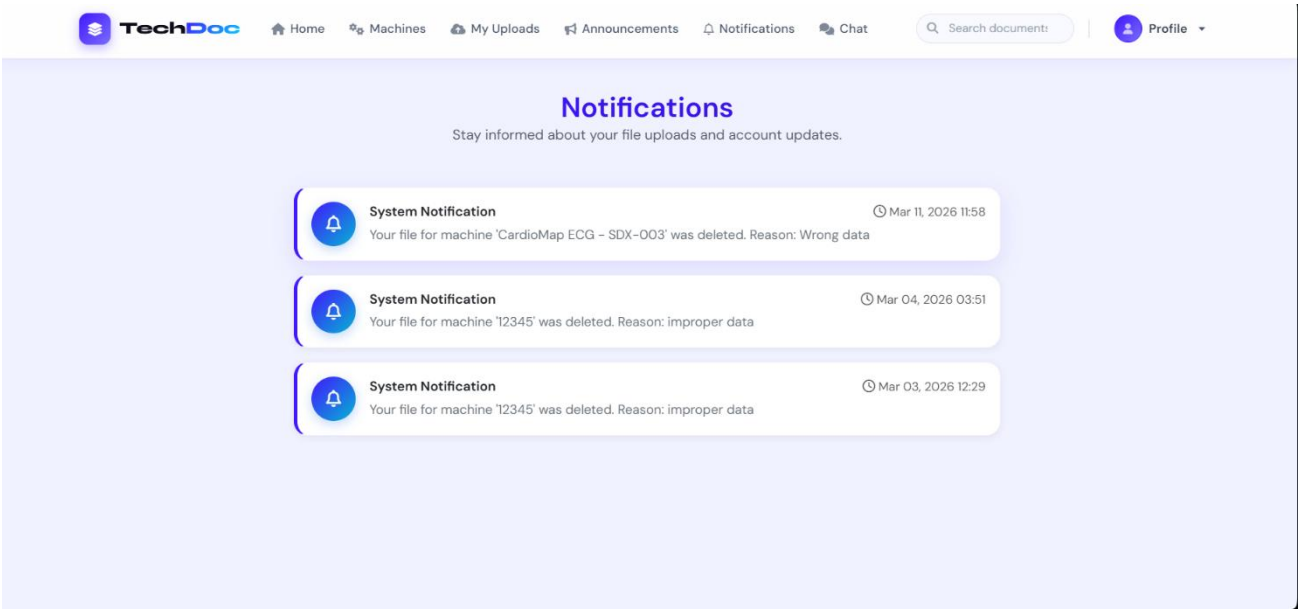
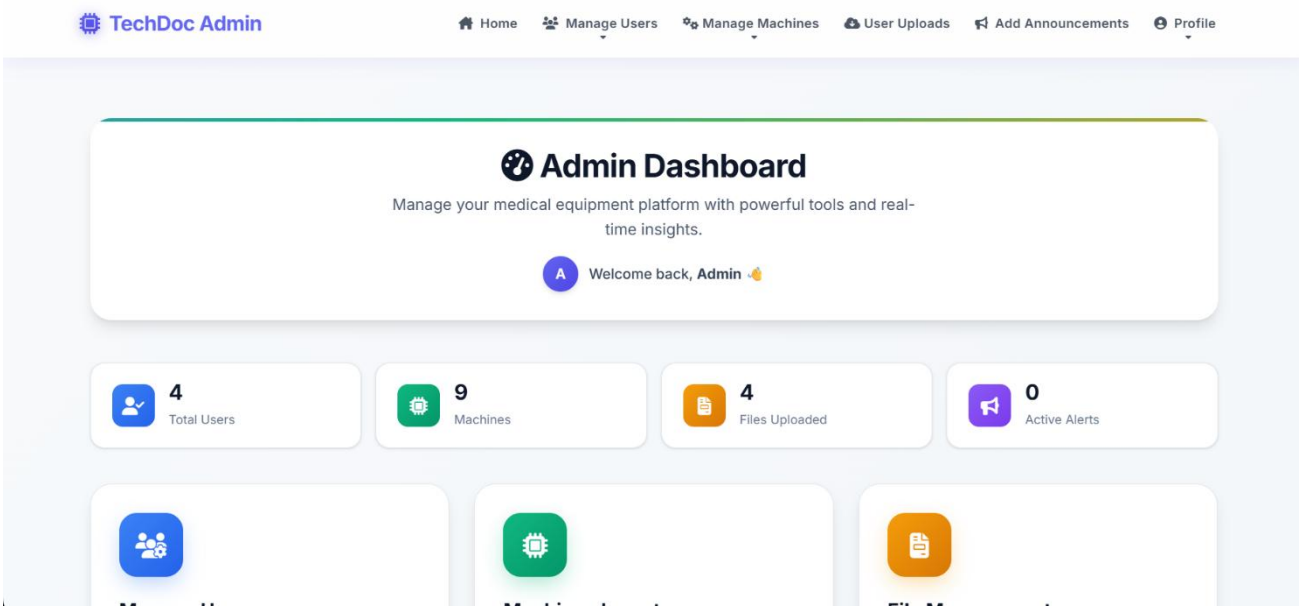


Figure 4.29: Notifications

The two images showcase the Notifications and Announcements pages of the TechDocHub system. The first image displays the Notifications page, where users can view important messages and alerts related to system activities, such as updates, file-related information, or other system-generated messages. This section helps users stay informed about important events and updates within the platform. The second image shows the Announcements page, where official notices posted by the administrator are displayed. These announcements may include system updates, information about new machines, or other important notices intended for all users, ensuring that users remain updated with the latest developments in the system.



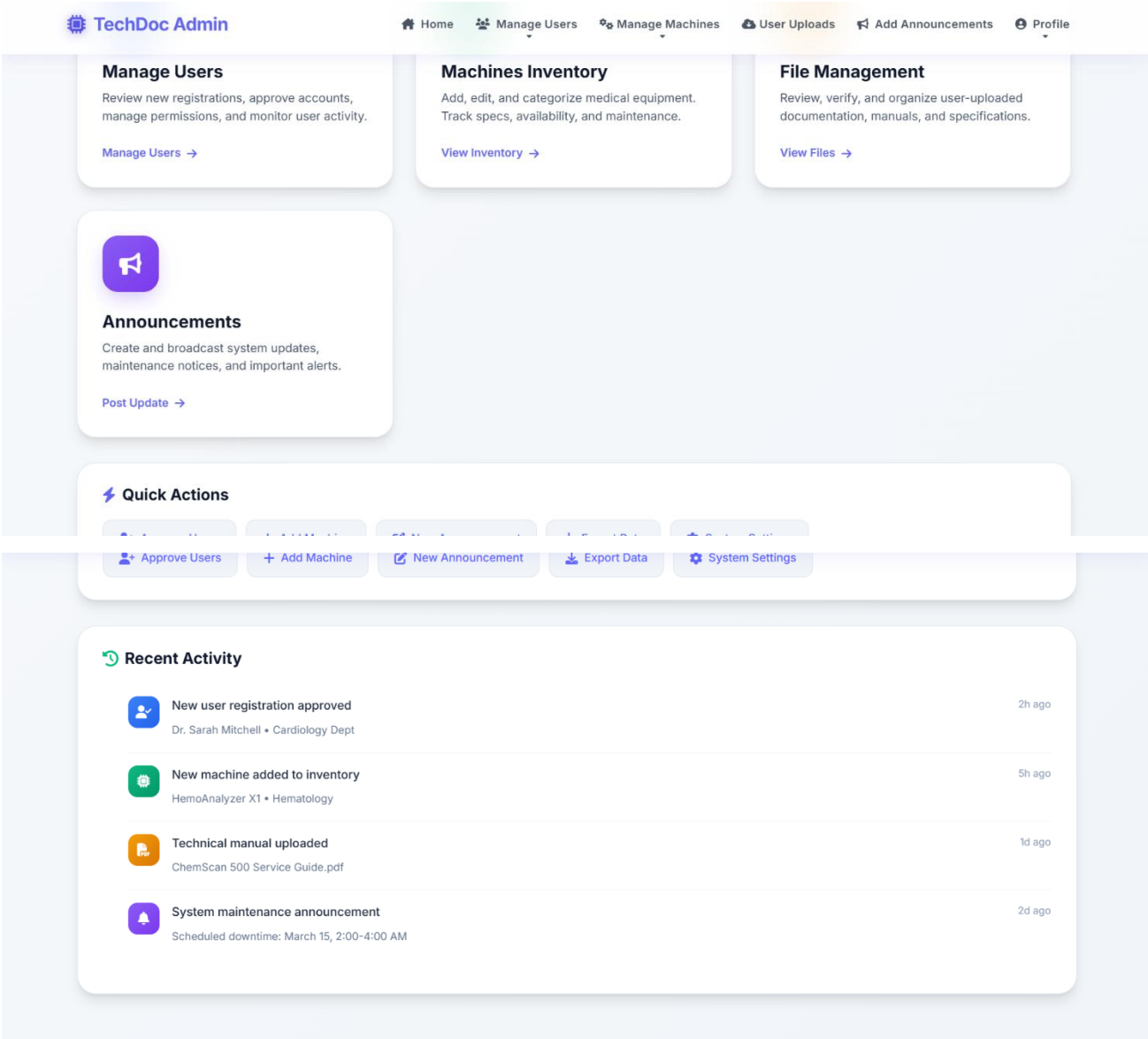


Figure 4.30: Admin Homepage

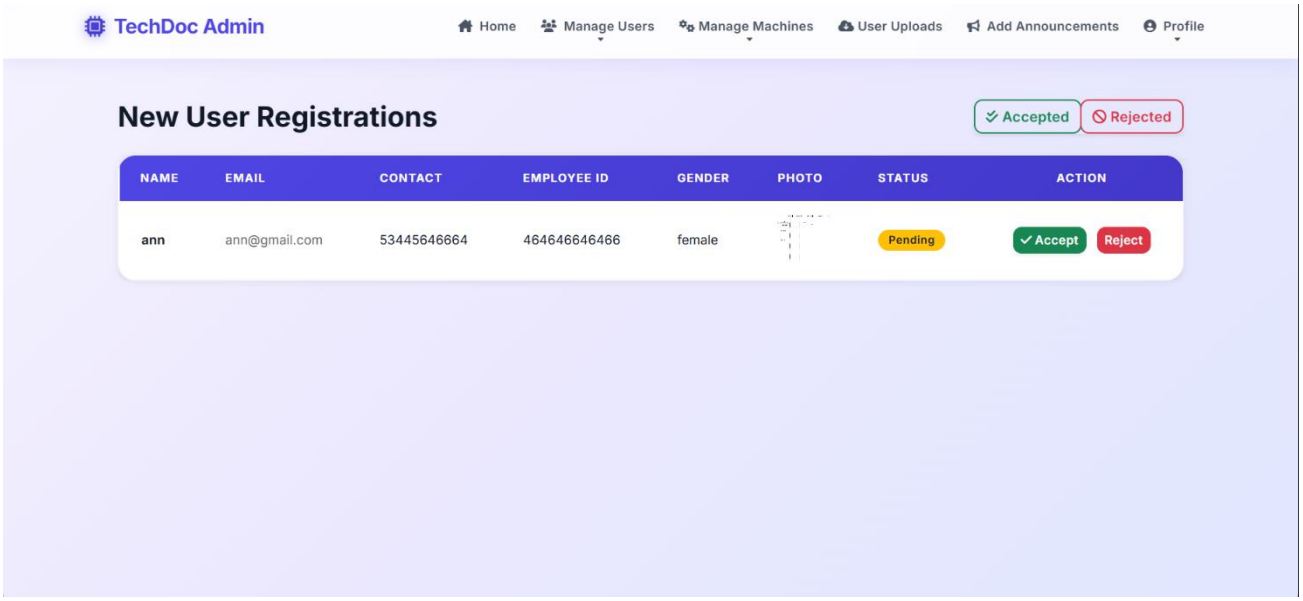


Figure 4.30: New User Approval

The two images showcase the Admin Home Page and New User Approval Page of the TechDocHub system. The first image displays the Admin Home Page, which provides the administrator with an overview of the system and quick access to important management features such as machine categories, machines, announcements, user machine files, and notifications. This page serves as the central dashboard from which the administrator can efficiently manage the platform. The second image shows the New User Approval Page, where the administrator can view newly registered users whose accounts are pending approval. From this page, the admin can review user details and approve or reject registrations, ensuring that only authorized users are allowed to access the system.

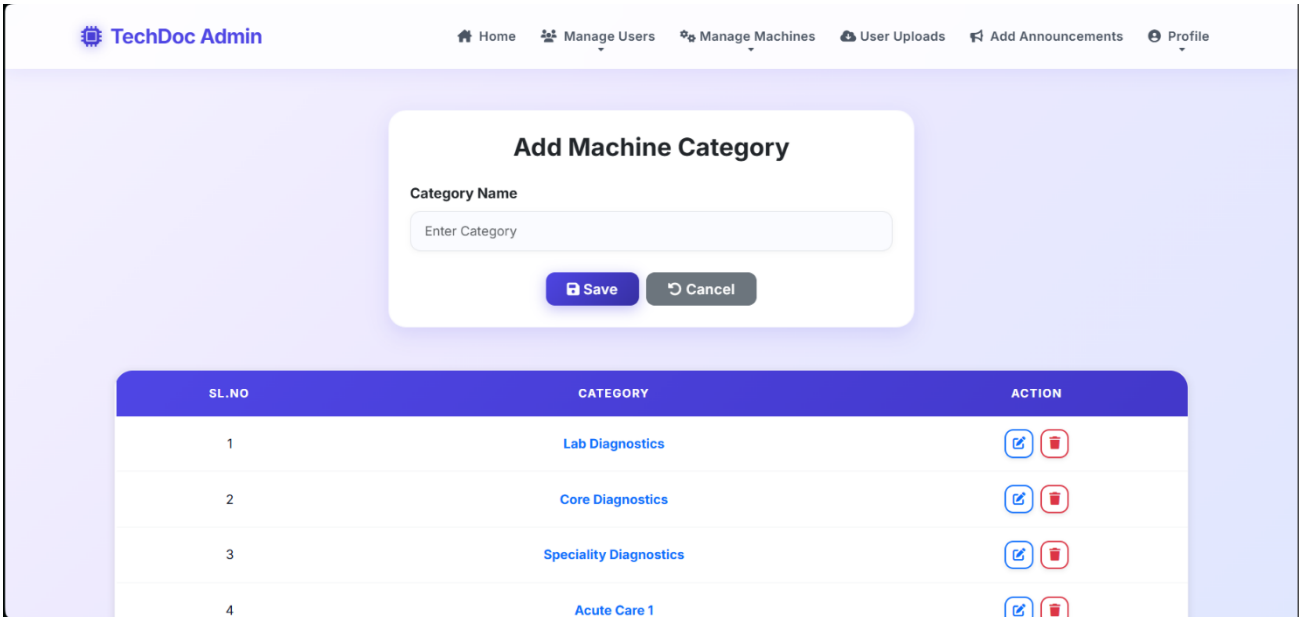


Figure 4.31: New Machine Category

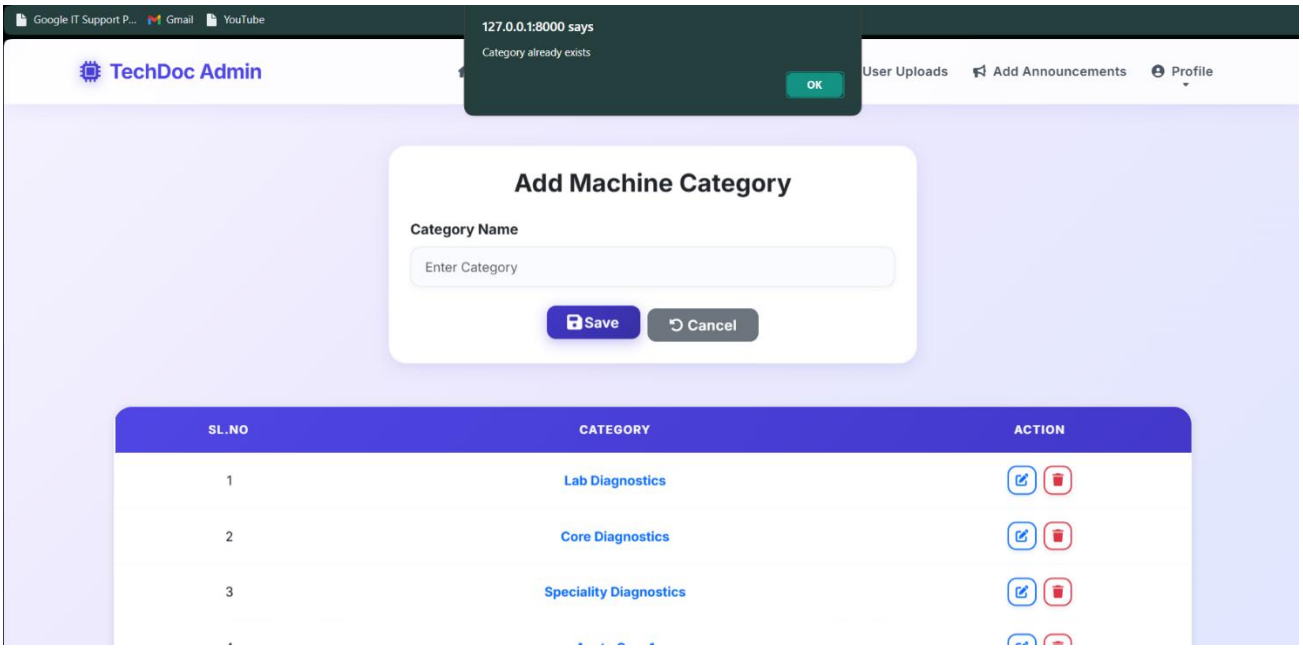


Figure 4.32: New Machine Category message alert

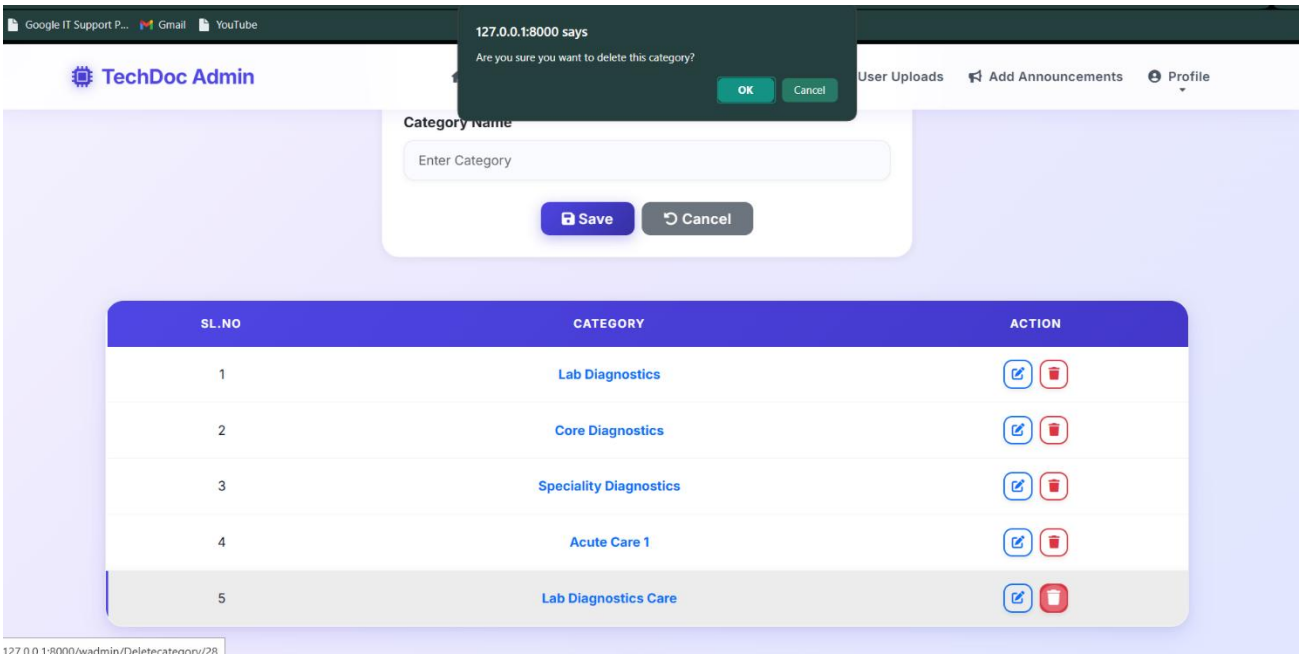


Figure 4.33: Deletion of Machine Category message alert

The three images showcase the New Machine Category page and system alert messages in the TechDocHub system. The first image displays the New Machine Category page, where the administrator can add a new category by entering the category name. This feature helps organize machines into different groups, making it easier for users to browse and find machine information. The second image shows a duplication message alert, which appears when the administrator attempts to add a machine category that already exists in the system. This alert prevents duplicate entries and ensures data consistency. The third image displays a delete confirmation alert, which is shown when the administrator attempts to delete a category or record. This alert asks for confirmation before deletion to prevent accidental removal of important data from the system.

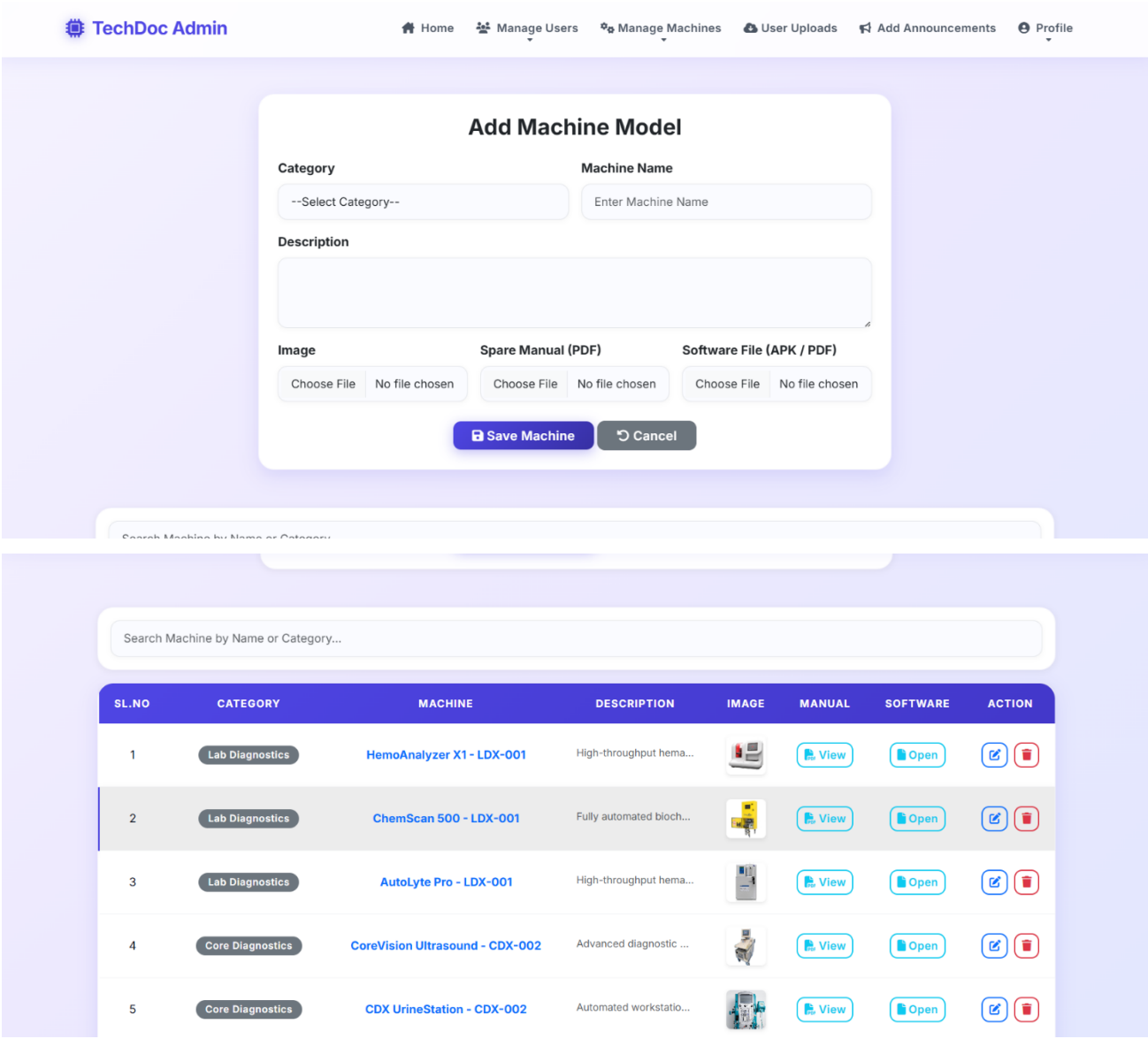


Figure 4.34: New Machine

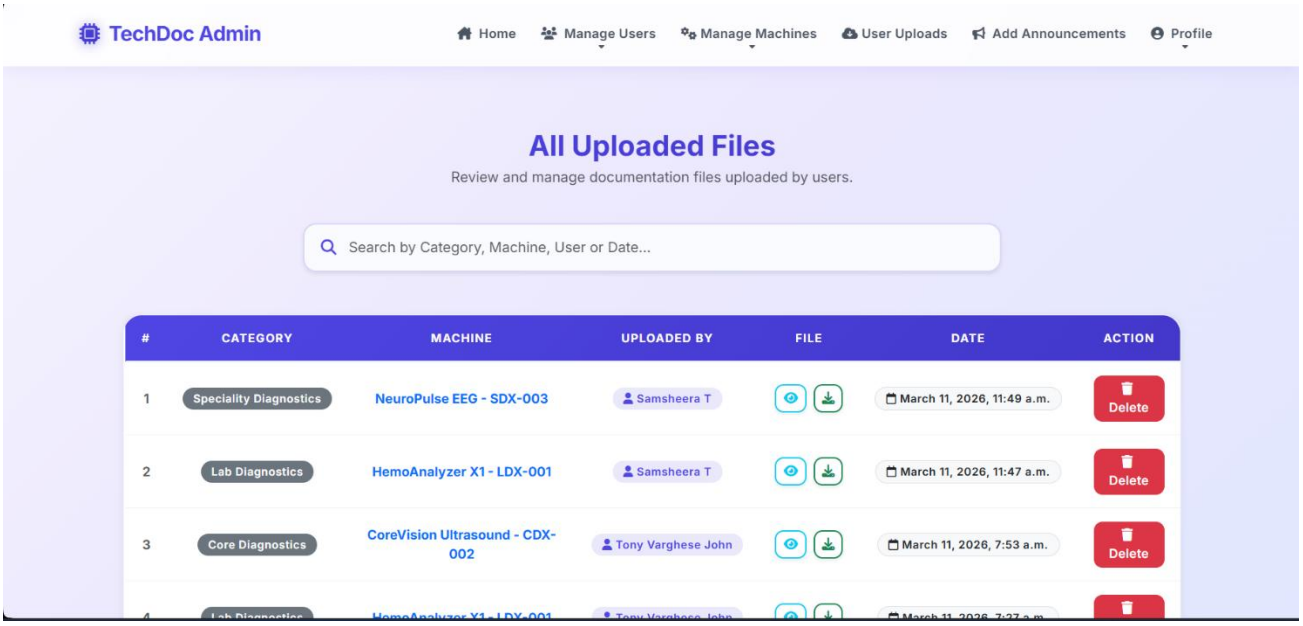


Figure 4.35: User Uploads

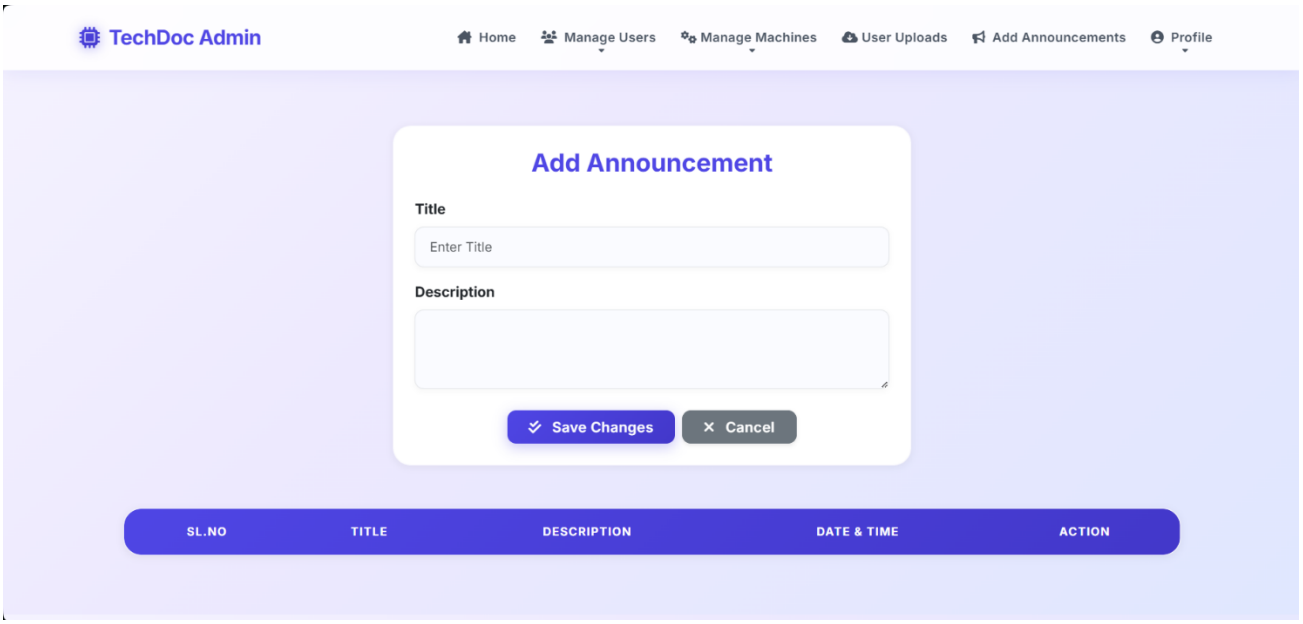


Figure 4.35: New Announcements

The three images showcase the New Machine, User Uploads, and New Announcement pages of the

TechDocHub system. The first image displays the New Machine page, where the administrator can add a new machine by entering details such as the machine category, machine name, description, image, spare manual, and related software files. This feature allows the administrator to maintain an organized collection of machines and their related resources within the platform. The second image shows the User Uploads page, which lists the files uploaded by users along with the associated machine details, enabling the administrator to view, manage, and delete uploaded files when necessary. The third image presents the New Announcement page, where the administrator can create and publish announcements by entering a title and description, allowing important updates and information to be communicated to all users of the system.

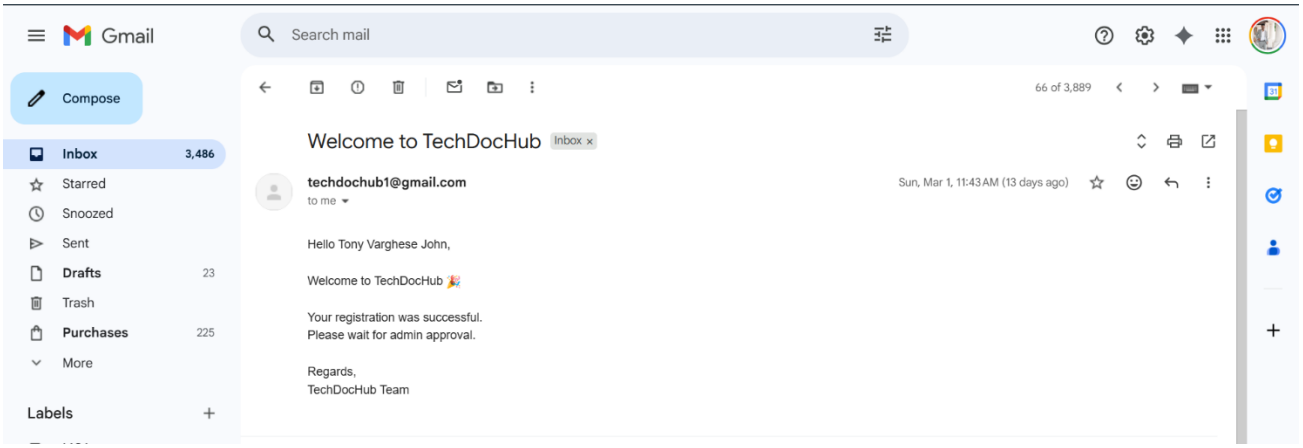


Figure 4.35: Email message

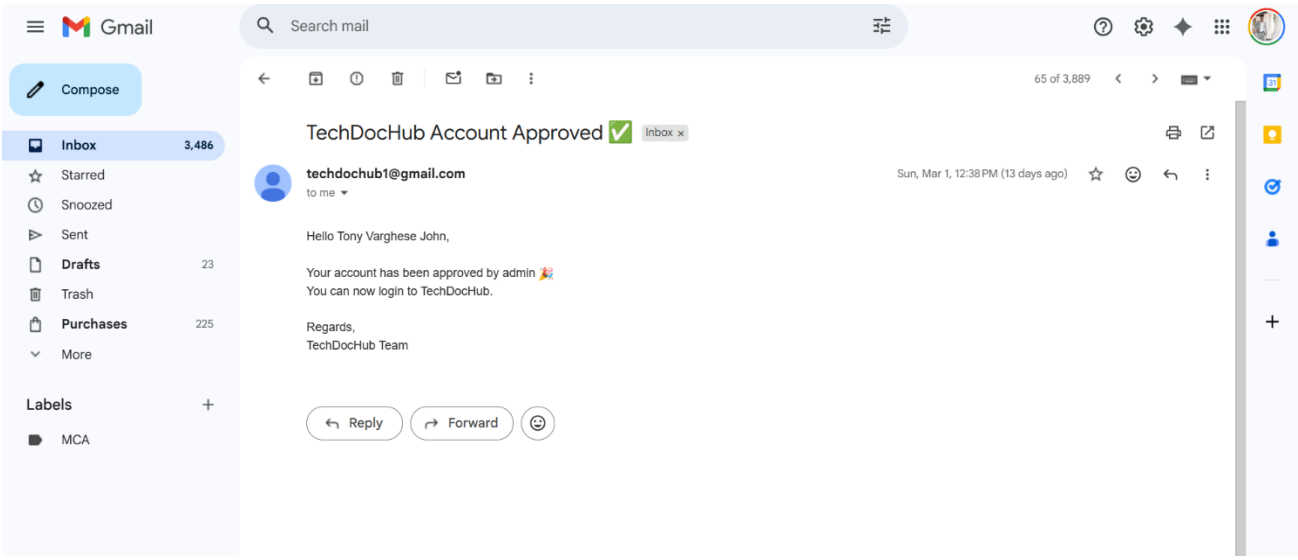


Figure 4.36: Email message

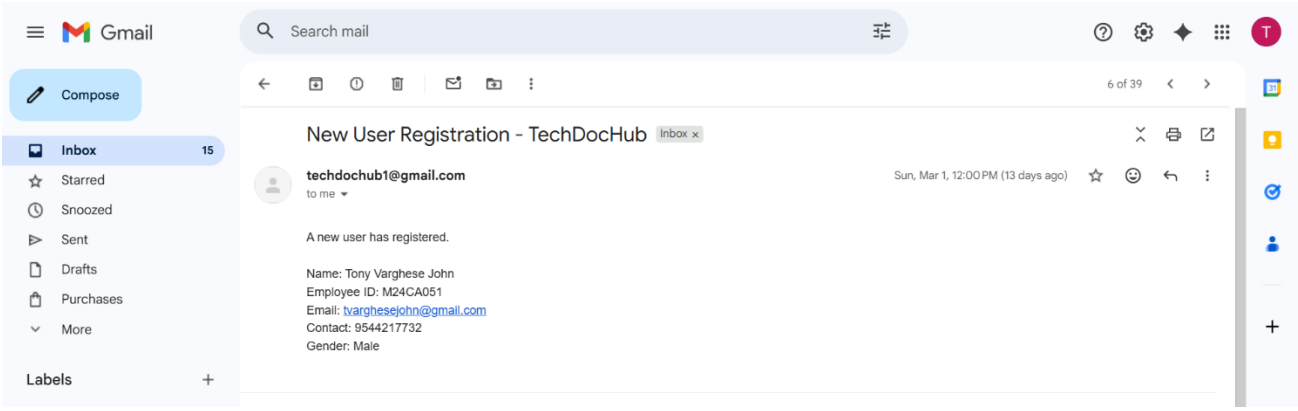


Figure 4.37: Email message

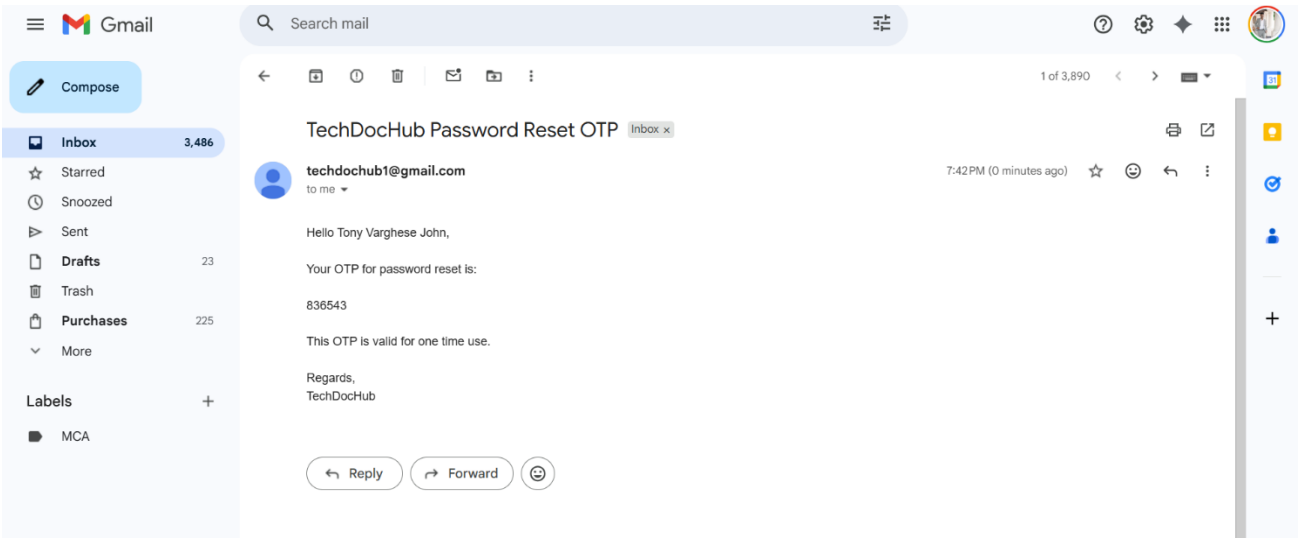


Figure 4.38: Email message

The four images showcase the Email Notification System of the TechDocHub platform, which is used to send automated emails to users for different system activities. The first image shows the user registration email, which is sent to confirm that a user has successfully registered in the system. The second image displays the account approval email, which is sent when the administrator approves a new user, allowing them to log in and access the platform. The third image presents the notification email, which informs users about important updates or announcements within the system. The fourth image shows another system-generated email alert, which may notify users about system activities such as file uploads or other important updates. These automated emails help keep users informed and improve communication between the system and its users.